The Microbe 9 (2025) 100624

Contents lists available at ScienceDirect

The Microbe

journal homepage: www.sciencedirect.com/journal/the-microbe

A graph theoretic and optimal control of bovine tuberculosis

Sylas Oswald [a,*], Theresia Crispin Marijani [a], Goodluck Mika Mlay [a], Winifrida Kidima [b]

[a] Department of Mathematics, University of Dar es Salaam, P.O Box 35062, Dar es Salaam, Tanzania
[b] Department of Zoology and Wildlife Ecology, University of Dar es Salaam, P.O Box 35061, Dar es Salaam, Tanzania

ARTICLE INFO

Keywords:
Mycobacterium bovis
Digraph
Self-loop
Averted
Hamiltonian function

ABSTRACT

Bovine tuberculosis (bTB), caused by *Mycobacterium bovis*, is a chronic zoonotic disease that poses significant risks to both public health and livestock productivity, particularly in developing countries such as Tanzania. In this study, we utilized human and cattle data collected between 2016 and 2019 to develop and analyse a deterministic compartmental model investigating the transmission dynamics of bTB in interacting human and cattle populations. The model incorporates three time-dependent control measures public health education, treatment, and vaccination implemented individually and in combination, with the dual objective of reducing infection prevalence and minimizing implementation costs.

The effective reproduction number, $\mathcal{R}_e$, was derived using a graph-theoretic Gaussian elimination approach. Analytical results demonstrate that the disease-free equilibrium is locally asymptotically stable when $\mathcal{R}_e < 1$, whereas an endemic equilibrium exists and is globally stable when $\mathcal{R}_e > 1$. Optimal control conditions were established through Pontryagin's Maximum Principle, and numerical simulations were performed using a forward–backward sweep algorithm combined with fourth-order Runge–Kutta schemes.

Simulation outcomes indicate that combined interventions particularly the integration of public health education, treatment, and vaccination substantially reduce bTB transmission, resulting in a marked decline in infection levels across both human and cattle populations. Cost-effectiveness analysis based on the Incremental Cost-Effectiveness Ratio (ICER) further reveals that vaccination alone represents the most economically viable strategy, especially in resource-constrained settings. Overall, these findings highlight that while vaccination is the most cost-effective single intervention, a comprehensive strategy combining education, treatment, and vaccination offers the greatest reduction in transmission, supporting long-term goals for bTB control and eventual eradication.

1. Introduction

Bovine tuberculosis (bTB) is a contagious and potentially fatal zoonotic disease caused by the bacterium *Mycobacterium bovis* (*M. bovis*). While it primarily affects cattle, *M. bovis* can infect a wide range of mammals, including humans, goats, pigs, cats, and dogs (Al-Obeidi and Al-Azzawi, 2022; Ekwem et al., 2021).

In Tanzania, the burden of zoonotic diseases is increasingly recognized by the Ministry of Livestock and Fisheries (MoLF) as a significant challenge to both public health and livestock productivity. Recent government–public reports indicate that approximately 60% of all outbreak-causing diseases originate from animals, underlining the scale of the issue at the human–animal interface (Guardian Reporter, 2024). This statistic conveys that a large majority of disease events have animal origins, prompting national campaigns such as "Holela Holela Itakukosti" launched by MoLF in partnership with USAID to promote responsible animal handling, livestock vaccination, and public awareness of zoonotic risk (Food and Agriculture Organization of the United Nations (FAO), 2024).

Despite this recognition, there remains a striking gap in context-specific research and modelling focused on zoonotic transmission dynamics in Tanzania. For example, a recent study in the northern regions found that in a set of 98 reported disease outbreaks, 63 (64.3%) were zoonotic in nature, illustrating how dominant zoonoses are among disease events in Tanzanian livestock systems (Giliard, 2025). Coupled with Tanzania's adoption of a national One Health strategic plan that notes 60%–80% of human pathogens originate from animals (Prime Minister's Office – The United Republic of Tanzania, 2022), the need for integrated research becomes clear.

* Corresponding author.
E-mail address: sylasossy2023@gmail.com (S. Oswald).

https://doi.org/10.1016/j.microb.2025.100624
Received 16 August 2025; Received in revised form 9 November 2025; Accepted 19 November 2025
Available online 26 November 2025
2950-1946/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

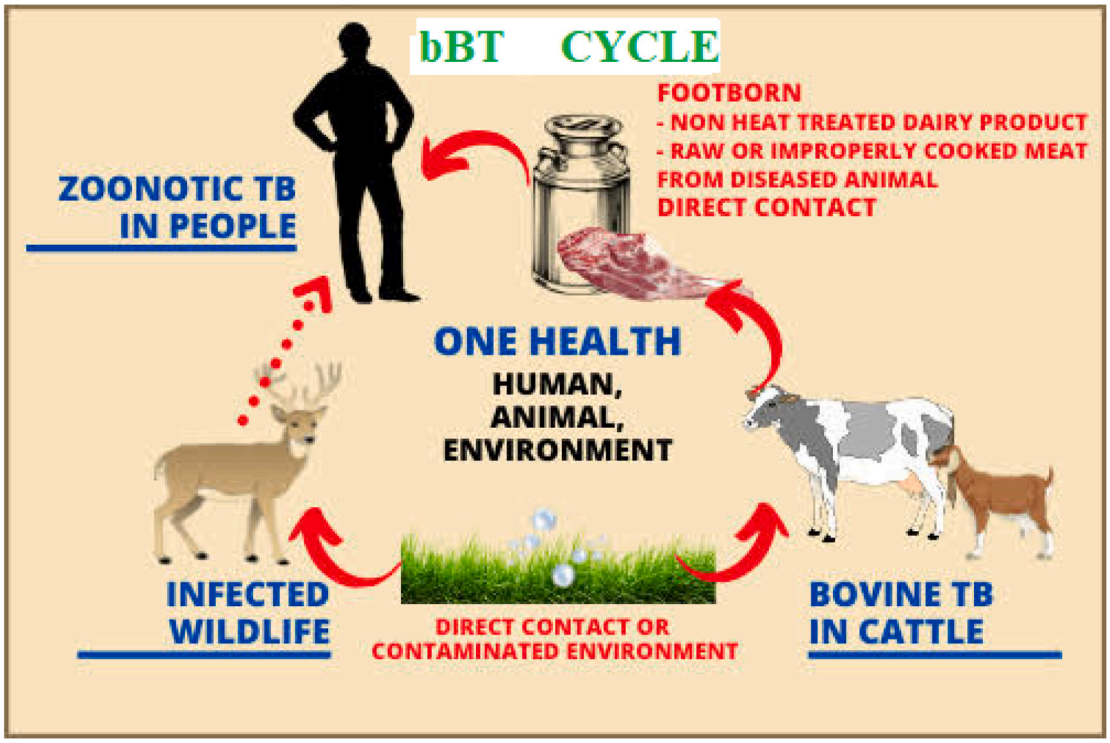


Fig. 1. Bovine tuberculosis transmission dynamics.

Transmission occurs mainly through inhalation of respiratory droplets from infected animals, although indirect transmission via ingestion or contact with contaminated materials such as sputum, pus, urine, faeces, or other excreta also contributes to the spread (Ekwem et al., 2021; Srinivasan et al., 2021).

The bacterium has been isolated from soil, animal waste, hay, and pasture, highlighting the challenge of environmental decontamination.

Humans are particularly at risk through consumption of unpasteurized dairy products or direct contact with infected animals, especially during slaughter or hunting activities. Although direct airborne transmission from animals to humans is uncommon, person-to-person transmission can occur via respiratory droplets. Occupational groups such as dairy farmers, ranchers, abattoir workers, butchers, and hunters are especially vulnerable (Ekwem et al., 2021; Mwenyeheri et al., 2014).

Clinical manifestations in cattle include reduced productivity, weight loss, general weakness, respiratory distress, and occasionally gastrointestinal symptoms. Infected cattle may remain asymptomatic for extended periods, with latent infections becoming active under stress or ageing (Ekwem et al., 2021; Oswald Sylas et al., 2024a).

In humans, bTB symptoms are similar to those of human TB, including weight loss, fever, fatigue, night sweats, and productive cough, with severe cases potentially affecting the lymph nodes, joints, and spine (Aldila et al., 2019; Rossi et al., 2015).

The environmental persistence of *M. bovis* significantly complicates control efforts. The bacterium can survive in aerosolized form for several hours and persist in moist, cool environments for extended periods from days to months facilitating indirect transmission (Al-Obeidi and Al-Azzawi, 2022; Centers for Disease Control and Prevention, 2012).

Despite its serious implications, bTB remains a neglected zoonotic disease in much of Africa, including Tanzania. Control efforts are hindered by weak surveillance, limited public awareness, and insufficient intervention strategies. The disease is endemic in several Tanzanian regions, including Arusha, Kilimanjaro, Manyara, Morogoro, and Kibaha, with prevalence estimates ranging from 0.2% to 13.3%, often associated with cattle density and pastoral practices (Durnez et al., 2009; Karimuribo et al., 2007; Mdegela et al., 2004; Sabini et al., 2019). Fig. 1 illustrates the primary direct and indirect transmission routes of *M. bovis* between cattle and humans.

Several mathematical models have been developed to study bovine tuberculosis (bTB) dynamics, particularly in African contexts. Aldila et al. (2019) proposed a deterministic model incorporating three transmission pathways cattle-to-cattle, human-to-human, and cattle-to-human and found that cattle quarantine and mask use significantly reduced the basic reproduction number ($\mathcal{R}_0$). However, the model did not include other key interventions such as vaccination, treatment, or public health education.

Building on this foundation, vaccination-based models have been explored in several studies. For instance, Rossi et al. (2015) demonstrated that vaccination alone can effectively reduce infection rates, while combining vaccination with public health education and treatment further lowered $\mathcal{R}_0$ below unity, even though treatment was not dynamically modelled. Similarly, Diallo et al. (2023) introduced a fractional-order model that incorporated vaccination and environmental transmission, but behavioural interventions, such as education, were not considered. Other studies, including Liu et al. (2016), highlighted the effectiveness of slaughter and quarantine in reducing cattle infections, consistent with field observations in Tanzania's Ngorongoro district, where high bTB prevalence is associated with semi-nomadic pastoral practices. Researchers such as Katale et al. (2012), Mlay and Hugo (2021) have accordingly recommended public health education for pastoralist communities as an essential mitigation measure.

In recent years, a growing number of studies have expanded the modelling of both tuberculosis (TB) and bovine tuberculosis (bTB), integrating new mathematical approaches and control strategies. For example, Sabini et al. (2020) developed a deterministic model of zoonotic bTB transmission

between cattle and humans in Tanzania, accounting for direct and indirect transmission routes but excluding control measures such as vaccination or education. More recently, Diallo et al. (2023, 2024) formulated fractional-order bTB models that incorporated vaccination and environmental contamination, yet did not include human–animal behavioural interactions or time-dependent controls. Similarly, Youssouf et al. (2025) explored bTB transmission in mixed-species herds in Chad, emphasizing environmental persistence but focusing solely on animal populations.

Beyond bTB, several human TB models have considered broader intervention frameworks that integrate biomedical and behavioural components. For instance, Simorangkir et al. (2021) analysed a TB model with vaccination and treatment, while Ullah et al. (2022) incorporated public health education and hospital treatment as control measures. More comprehensive studies, such as Zhang and Yang (2024), examined optimal control strategies combining vaccination, treatment, and environmental contamination, whereas Rizwan et al. (2023) introduced fractional derivatives to capture memory effects in treatment dynamics. These works illustrate the methodological evolution of TB modelling but also underscore a limitation, most models focus exclusively on either human or animal systems, rarely integrating cross-species transmission or behavioural components.

Cross-species dynamics and ecological interfaces further complicate bTB control efforts. Hassan et al. (2014b) modelled bTB transmission in a human–buffalo system, revealing complex dynamics such as backward bifurcation. Similarly, Clifford et al. (2022) investigated bTB spillover risks in Tanzania's Ruaha ecosystem, while Katale et al. (2017) provided important prevalence data from cattle and wildlife in the Serengeti, though without formal modelling of intervention strategies. Environmental persistence of *M. bovis* was also considered in the model by Diallo et al. (2023), which included an environmental compartment but did not examine sanitation or education-based controls. In contrast, Yahyaoui-Azami et al. (2020) focused on test-and-cull strategies in North Africa but likewise omitted behavioural or public health outreach components. Although Bronsvoort et al. (2010) modelled cattle vaccination in Cameroon and found it to be cost-effective, integrated human–animal vaccination modelling remains rare. A broader review by Katale et al. (2024) emphasized this gap, highlighting the lack of context-specific models that integrate biomedical and behavioural interventions within a single framework.

Despite these advancements, a clear research gap remains. Most existing models concentrate on a single type of intervention vaccination, treatment, or education, restrict their scope to either human or animal hosts. Very few studies have developed a fully integrated zoonotic TB model that simultaneously captures cattle–human transmission, environmental persistence, and multiple intervention strategies under a unified, time-dependent control framework. Moreover, while human TB models increasingly incorporate behavioural interventions such as education campaigns (Ullah et al., 2022), such approaches are largely absent from bTB frameworks, particularly those tailored to endemic settings in sub-Saharan Africa.

To address these limitations, the present study develops a dynamic model for bovine tuberculosis transmission that integrates three key interventions, public health education campaigns, treatment, and vaccination as time-dependent control measures. The model is tailored to the Tanzanian context, where bTB remains endemic, and aims to assess the individual and combined effects of these interventions on disease dynamics. This integrated framework provides a novel and policy-relevant approach to bTB control in resource-limited settings, bridging the gap between biomedical and behavioural strategies.

2. Model formulation

The model developed in this study considers two interacting populations humans and cattle that regularly come into contact within a ranching system, where only cattle are confined. At any time t, the total human population, denoted by $N_h(t)$, is subdivided into five epidemiological compartments: susceptible non-educated individuals $S_{nh}(t)$, susceptible educated individuals $S_{eh}(t)$, exposed individuals $E_h(t)$, infectious individuals $I_h(t)$, and recovered individuals $R_h(t)$. Thus, the total human population is expressed as:

$$N_h(t) = S_{nh}(t) + S_{eh}(t) + E_h(t) + I_h(t) + R_h(t).$$

Humans enter the susceptible non-educated class at a constant recruitment rate π_h. Public health education is administered exclusively to individuals in the S_{nh} class at a rate ψ, aiming to reduce disease transmissibility. However, the protective effect of education is assumed to be temporary. Susceptible non-educated individuals may acquire infection through the consumption of contaminated cattle products, inhalation of aerosols from infected cattle, or human-to-human contact, subsequently progressing to the exposed class. The force of infection for susceptible non-educated humans, denoted by φ_1, is given by:

$$\varphi_1 = (1-\theta)\left(a_1\frac{I_c}{N_h} + a_2\frac{I_h}{N_h}\right),$$

where a_1 and a_2 represent the probabilities of transmission from infectious cattle and infectious humans, respectively, and $\theta \in [0,1]$ denotes the efficacy of the public health education campaign.

Educated susceptible individuals may also contract the disease through contact with contaminated cattle products and aerosols from infected cattle. The corresponding force of infection, φ_2, is defined as:

$$\varphi_2 = a\theta\frac{I_c}{N_h},$$

where a is the probability of transmission from infectious cattle to educated susceptible humans.

Both susceptible classes may leave their compartments due to natural death at a rate μ_h. Exposed individuals progress to the infectious class or recover at a rate ρ_h, with additional exits due to natural mortality μ_h or disease-induced mortality α_h. Recovered individuals are assumed not to acquire permanent immunity and may return to the exposed class at a rate τ_h or exit through natural death at rate μ_h.

The cattle population at time t, denoted by $N_c(t)$, is classified into four compartments: susceptible non-vaccinated $S_{nv}(t)$, susceptible vaccinated $S_v(t)$, exposed $E_c(t)$, and infectious $I_c(t)$. Therefore,

$$N_c(t) = S_{nv}(t) + S_v(t) + E_c(t) + I_c(t).$$

Cattle enter the non-vaccinated susceptible class at a constant rate π_c. Healthy susceptible cattle are vaccinated at a rate ω, reducing transmission of *M. bovis* from environmental sources and other cattle.

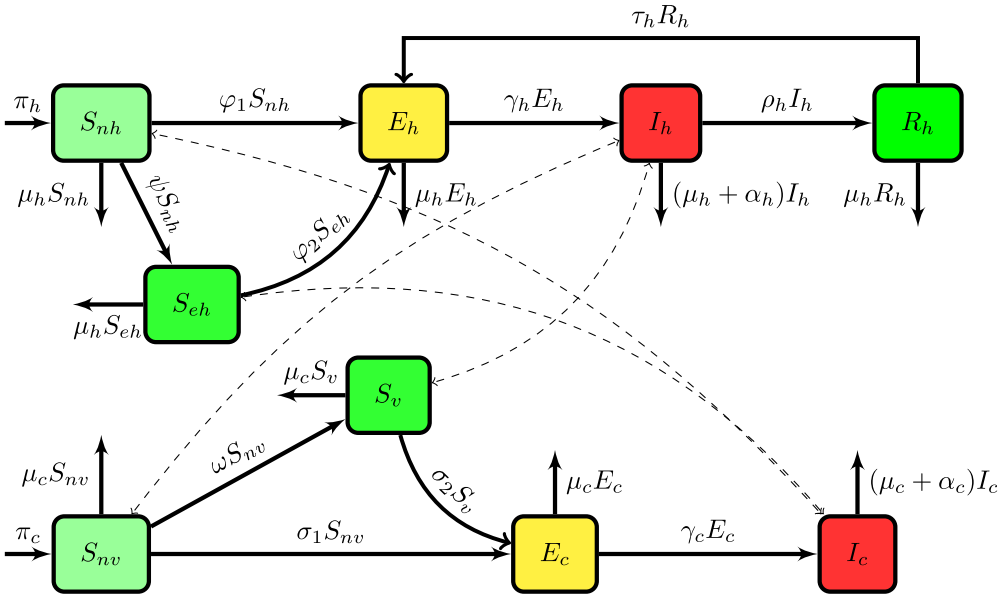


Fig. 2. Schematic flow diagram for the dynamics of bTB disease in the presence of some intervention strategies.

Susceptible non-vaccinated cattle may acquire infection via inter-cattle transmission, environmental exposure, or contact with infectious humans, progressing to the exposed class at a rate σ_1. The corresponding force of infection is:

$$\sigma_1 = (1-\lambda)\left(\varepsilon_1 \frac{I_h}{N_c} + \varepsilon_2 \frac{I_c}{N_c}\right),$$

where ε_1 is the probability of infection from humans, ε_2 captures infection from the environment including contaminated grazing areas and direct contact (e.g., nose-to-nose, aerosols) and $\lambda \in [0,1]$ represents vaccine efficacy. If no cattle are vaccinated, then $\lambda = 0$.

Vaccinated cattle may still acquire infection due to incomplete vaccine protection, with the force of infection defined as:

$$\sigma_2 = \varepsilon\lambda \frac{I_h}{N_c},$$

where ε denotes the probability of infectious human-to-cattle transmission despite vaccination.

Both vaccinated and non-vaccinated susceptible cattle exit their compartments through natural death at a rate μ_c. Exposed cattle either progress to the infectious class at rate γ_c or die naturally at rate μ_c. Infectious cattle leave the compartment through natural mortality μ_c or disease-induced mortality α_c.

Based on the underlying assumptions and epidemiological relationships summarized in Table 1, and as illustrated schematically in Fig. 2, the transmission dynamics of bovine tuberculosis under the influence of three key intervention strategies, public health education, treatment, and vaccination are formulated through a system of nine coupled ordinary differential equations (ODEs). This model captures the interaction between human and cattle populations, incorporating disease progression, control interventions, and demographic processes to simulate the transmission pathway and evaluate intervention impacts.

$$\begin{cases}
\frac{dS_{nh}}{dt} &= \pi_h - (\mu_h + \psi + \varphi_1)S_{nh},\\
\frac{dS_{eh}}{dt} &= \psi S_{nh} - (\mu_h + \varphi_2)S_{eh},\\
\frac{dE_h}{dt} &= \varphi_1 S_{nh} + \varphi_2 S_{eh} - (\mu_h + \gamma_h)E_h + \tau_h R_h,\\
\frac{dI_h}{dt} &= \gamma_h E_h - (\mu_h + \alpha_h + \rho_h)I_h,\\
\frac{dR_h}{dt} &= \rho_h I_h - (\mu_h + \tau_h)R_h,\\
\frac{dS_{nv}}{dt} &= \pi_c - (\mu_c + \omega + \sigma_1)S_{nv},\\
\frac{dS_v}{dt} &= \omega S_{nv} - (\mu_c + \sigma_2)S_v,\\
\frac{dE_c}{dt} &= \sigma_1 S_{nv} + \sigma_2 S_v - (\mu_c + \gamma_c)E_c,\\
\frac{dI_c}{dt} &= \gamma_c E_c - (\mu_c + \alpha_c)I_c,
\end{cases} \tag{1}$$

where $\varphi_1 = (1-\theta)\left(a_1 \frac{I_c}{N_h} + a_2 \frac{I_h}{N_h}\right)$, $\varphi_2 = a\theta \frac{I_c}{N_h}$, $\sigma_1 = (1-\lambda)\left(\varepsilon_1 \frac{I_h}{N_c} + \varepsilon_2 \frac{I_c}{N_c}\right)$, $\sigma_2 = \varepsilon\lambda \frac{I_h}{N_c}$.

Table 1
Model parameters and their epidemiological interpretations.

Parameter	Epidemiological interpretation	Value	Source
π_h	Recruitment rate into the human population	36	Liu et al. (2016)
π_c	Recruitment rate into the cattle population	200	Liu et al. (2016)
a_1	Probability of transmission from infectious cattle to susceptible non-educated humans	0.55	Hassan et al. (2014a)
a_2	Probability of transmission from contaminated environment or via inter-human contact	0.35	Shirima et al. (2020)
a	Probability of transmission from infectious cattle to susceptible educated humans	0.65	Liu et al. (2016)
ε_1	Probability of transmission from infectious humans to susceptible cattle	0.803	Liu et al. (2016)
ε_2	Probability of cattle infection via environment or inter-cattle contact	0.908	Calculated
ε	Probability of transmission from infectious humans to vaccinated susceptible cattle	0.32	Liu et al. (2016)
ρ_h	Recovery rate of infectious humans due to treatment	0.72	Fitted
τ_h	Rate of waning immunity among recovered humans, returning to latent class	0.01	Fitted
γ_h	Progression rate of exposed humans to infectious stage	0.48	Fitted
γ_c	Progression rate of exposed cattle to infectious stage	0.63	Fitted
μ_h	Natural death rate in humans	0.013	Calculated
μ_c	Natural death rate in cattle	0.13	Calculated
α_h	Disease-induced mortality rate in humans	0.43	Fitted
α_c	Disease-induced mortality rate in cattle	0.62	Fitted
θ	Effectiveness of public health education campaigns	0.75	Fitted
λ	Effectiveness of vaccination among cattle	0.35	Fitted
ψ	Per capita rate of education among non-educated humans	0.745	Fitted
ω	Per capita rate of vaccination among susceptible cattle	0.67	Oswald Sylas (2023)

2.1. Positivity of the solutions

For the model (1) to be epidemiologically meaningful and well-posed, it is necessary to prove that all the state variables are non-negative $\forall t \geq 0$.

Lemma 2.1. *Let* $\{S_{nh}(0) > 0,\ S_{eh}(0) > 0,\ E_h(0) \geq 0,\ I_h(0) \geq 0,\ R_h(0) \geq 0,\ S_{nv}(0) > 0,\ S_v(0) > 0,\ E_c(0) \geq 0$ *and* $I_c(0) \geq 0\ \}$ *of the model system* (1) *are satisfied then the solutions* $\{S_{nh},\ S_{eh},\ E_h,\ I_h,\ R_h,\ S_{nv},\ S_v,\ E_c,\ I_c\ \}$ *are non-negative for all* $t \geq 0$ *in* Ω.

Proof. From the first equation of system (1),

$$\frac{dS_{nh}}{dt} = \pi_h - (\mu_h + \psi + \varphi_1)S_{nh} > -\left(\mu_h + \psi + \varphi_1\right) S_{nh}.$$

In the absence of human recruitment π_h it follows that $\implies \dfrac{dS_{nh}}{S_{nh}} > -\left(\mu_h + \psi + \varphi_1\right)\, dt.$

Upon integration gives,

$$\frac{dS_{nh}}{S_{nh}} > -\left(\mu_h + \psi + \varphi_1\right) dt \implies \int_0^t \frac{dS_{nh}}{S_{nh}} ds > -\int_0^t \left(\mu_h + \psi + \varphi_1\right) ds.$$

$$\ln S_{nh}(s)|_0^t > -\int_0^t (\mu_h + \psi + \varphi_1) ds \implies \ln S_{nh}(t) - \ln S_{nh}(0) > -\int_0^t (\mu_h + \psi + \varphi_1) ds.$$

$$\log_e\left(\frac{S_{nh}(t)}{S_{nh}(0)}\right) > -\int_0^t (\mu_h + \psi + \varphi_1) ds \implies \left(\frac{S_{nh}(t)}{S_{nh}(0)}\right) > \exp\left(-\int_0^t (\mu_h + \psi + \varphi_1) ds\right)$$

$$S_{nh}(t) > S_{nh}(0)\exp\left(-\int_0^t (\mu_h + \psi + \varphi_1) ds\right) > 0.$$

In the same way, the remaining state variables give;

$$\begin{aligned}
S_{nh}(t) &> S_{nh}(0)\exp\left(-\int_0^t (\mu_h + \psi + \varphi_1) dt\right) \geq 0,\\
S_{eh}(t) &> S_{eh}(0)\exp\left(-\int_0^t (\mu_h + \varphi_2) dt\right) \geq 0,\\
E_h(t) &\geq E_h(0)\exp\left(-(\mu_h + \gamma_h)t\right) \geq 0,\\
I_h(t) &\geq I_h(0)\exp\left(-(\mu_h + \alpha_h + \rho_h)t\right) \geq 0,\\
R_h(t) &\geq R_h(0)\exp\left(-(\mu_h + \tau_h)t\right) \geq 0,\\
S_{nv}(t) &> S_{nv}(0)\exp\left(-\int_0^t (\mu_c + \omega + \sigma_1) dt\right) \geq 0,\\
S_v(t) &> S_v(0)\exp\left(-\int_0^t (\mu_c + \sigma_2) dt\right) \geq 0,\\
E_c(t) &\geq E_c(0)\exp\left(-(\mu_c + \gamma_c)t\right) \geq 0,\\
I_c(t) &\geq I_c(0)\exp\left(-(\mu_c + \alpha_c)t\right) \geq 0.
\end{aligned} \tag{2}$$

Therefore, the solution set $\{S_{nh},\ S_{eh},\ E_h,\ I_h,\ R_h,\ S_{nv},\ S_v,\ E_c,\ I_c\ \}$ of the model system (1) is non-negative for all $t \geq 0$ in Ω. □

2.2. Invariant region

Lemma 2.2. *The feasible region defined by* $\Omega = \Omega_h \cup \Omega_c \in R_+^5 \times R_+^4$ *where* $\Omega_h = \{S_{nh}, S_{eh}, E_h, I_h, R_h \in R_+^5 : S_{nh} + S_{eh} + E_h + I_h + R_h = N_h \leq \frac{\pi_h}{\mu_h}\}$ *and* $\Omega_c = \{S_{nv}, S_v, E_c, I_c \in R_+^4 : S_{nv} + S_v + E_c + I_c = N_c \leq \frac{\pi_c}{\mu_c}\}$, *is positively invariant and attracting with regard to the model system (1).*

Proof. Given $S_{nh}(0) \geq 0$, $S_{eh}(0) \geq 0$, $E_h(0) \geq 0$, $I_h(0) \geq 0$, $R_h(0) \geq 0$, $S_{nv}(0) \geq 0$, $S_v(0) \geq 0$, $E_c(0) \geq 0$ and $I_c(0) \geq 0$, it is sufficient to prove that $\{S_{nh}(t), S_{eh}(t), E_h(t), I_h(t), R_h(t), S_{nv}(t), S_v(t), E_c(t), I_c(t)\} \in R_+^9$ is bounded.

By summing equations for human population from model system (1) gives;

$$\frac{dN_h}{dt} = \pi_h - \mu_h N_h - \alpha_h I_h \leq \pi_h - \mu_h N_h$$

. Separating variables and applying anti-derivative of an integrating factor it gives;

$$\int \frac{d}{dt}\left(N_h(t)e^{\mu_h t}\right) dt \leq \int \pi_h e^{\mu_h t} dt \implies \int_0^t \frac{d}{ds}\left(N_h(s)e^{\mu_h s}\right) ds \leq \int_0^t \pi_h e^{\mu_h s} ds$$

$$N_h(s)e^{\mu_h s}\big|_0^t \leq \frac{\pi_h}{\mu_h} e^{\mu_h s}\big|_0^t \implies N_h(t)e^{\mu_h t} - N_h(0) \leq \frac{\pi_h}{\mu_h} e^{\mu_h t} - \frac{\pi_h}{\mu_h}.$$

$$N_h(t)e^{\mu_h t} \leq N_h(0) + \frac{\pi_h}{\mu_h} e^{\mu_h t} - \frac{\pi_h}{\mu_h} \implies N_h(t) \leq \frac{\pi_h}{\mu_h} - \left(\frac{\pi_h - \mu_h N_h(0)}{\mu_h}\right) e^{-\mu_h t}$$

$$N_h(t) \leq \frac{\pi_h}{\mu_h} - \left(\frac{\pi_h - \mu_h N_h(0)}{\mu_h}\right) e^{-\mu_h t}. \tag{3}$$

The same procedure can be used to prove that;

$$N_c(t) \leq \frac{\pi_c}{\mu_c} - \left(\frac{\pi_c - \mu_c N_c(0)}{\mu_c}\right) e^{-\mu_c t}. \tag{4}$$

Since the total population $N_h(t)$ as well as $N_c(t)$ is positive for all $t \geq 0$. It is well defined that,

$$\limsup_{t\to\infty} N_h(t) \leq \frac{\pi_h}{\mu_h} \quad \text{and} \quad \limsup_{t\to\infty} N_c(t) \leq \frac{\pi_c}{\mu_c}.$$

Therefore,

$$0 \leq N_h(t) \leq \frac{\pi_h}{\mu_h} \quad \text{and} \quad 0 \leq N_c(t) \leq \frac{\pi_c}{\mu_c}, \quad \text{for all} \quad t > 0. \tag{5}$$

Eq. (5) establishes that the proposed model system is epidemiologically valid, as it is mathematically well-posed and guarantees that all solutions to the bTB model, once initialized, remain within the feasible region. This confirms that the bTB model system (1) provides a meaningful epidemiological interpretation, particularly when all state variables are positive. As a result, the solution is constrained within bounds and attracts all trajectories in the domain Ω. Therefore, the dynamics of the bTB model system (1) can be effectively analysed within the domain Ω. □

2.3. Disease Free Equilibrium (DFE)

Disease-free equilibrium (DFE) of the bTB model system (1) is the point where there is no disease. Disease-free equilibrium point is obtained by setting $E_h = I_h = R_h = E_c = I_c = 0$ in all equations. Now, Let E_0 be disease-free equilibrium point. Therefore, the disease free equilibrium point (DFE) denoted by E_0 can be expressed as $E_0 = \left[\frac{\pi_h}{\mu_h + \psi}, \frac{\psi}{\mu_h}\left(\frac{\pi_h}{\psi + \mu_h}\right), 0, 0, 0, \frac{\pi_c}{\mu_c + \omega}, \frac{\omega}{\mu_c}\left(\frac{\pi_c}{\omega + \mu_c}\right), 0, 0\right]$.

2.3.1. The effective reproduction number

The effective reproduction number, $\mathcal{R}_e$, is a fundamental concept in epidemiology and medicine that represents the expected number of secondary infections generated by a single infected individual in a population that includes both susceptible and non-susceptible hosts (Castillo-Chavez et al., 2002; Oswald Sylas et al., 2024a; Castillo-Chavez and Song, 2004; Diekmann et al., 1990; Oswald et al., 2024b). It is a key metric for assessing the potential spread of infection and the effectiveness of control strategies (Oswald Sylas et al., 2025b; Naaly et al., 2024; Ndendya et al., 2024). Importantly, $\mathcal{R}_e$ acts as a threshold parameter indicating whether bTB transmission will decline or persist: if $\mathcal{R}_e \leq 1$, the infection will eventually die out, whereas $\mathcal{R}_e > 1$ implies sustained transmission and an increased bTB. To capture the complexities of disease spread, including the influence of prior infections, the model employs a Graph-Theoretic (GT) approach with a $\mathbb{GT}$-based Gaussian elimination method to compute $\mathcal{R}_e$, as detailed in de Camino-Beck et al. (2009), de Camino-Beck and Lewis (2008, 2007), Ndendya Jufren Zakayo et al. (2025g), Oswald Sylas et al. (2025a).

Theorem 1. *Let* $\mathcal{F}$ *be a non-negative, non-zero matrix, and* $\mathcal{M}$ *a non-singular M-matrix, under the assumption that* $\mathcal{F} - \mathcal{M}$ *is irreducible. Then, the effective reproduction number is given by* $\mathcal{R}_e = \rho(\mathcal{F}\mathcal{M}^{-1})$, *where* ρ *denotes the spectral radius and* $\mathcal{R}_e > 0$. *Furthermore,* $\mathcal{R}_e$ *corresponds to the reciprocal of the smallest positive root* ξ *of the equation* $\det(\mathcal{F}\xi - \mathcal{M}) = 0$.

Proof. Given the stated assumptions, the matrix product $\mathcal{F}\mathcal{M}^{-1}$ is non-negative and non-zero. According to Li and Schneider (2002), the principal submatrix corresponding to the non-zero rows of $\mathcal{F}$ is irreducible. As a result, the spectral radius $\xi = \rho(\mathcal{F}\mathcal{M}^{-1}) > 0$ represents the dominant eigenvalue of the characteristic equation $\det(\xi I - \mathcal{F}\mathcal{M}^{-1}) = 0$. Since $\mathcal{M}$ is non-singular and $\xi > 0$, this equation can be equivalently rewritten as $\det(\mathcal{F}\xi^{-1} - \mathcal{M}) = 0$, where $\mathcal{M} - \mathcal{F}\xi^{-1}$ becomes a singular M-matrix. Thus, the smallest positive solution to this equation is ξ^{-1}, and the effective reproduction number is given by $\mathcal{R}_e = \rho(\mathcal{F}\mathcal{M}^{-1})$. □

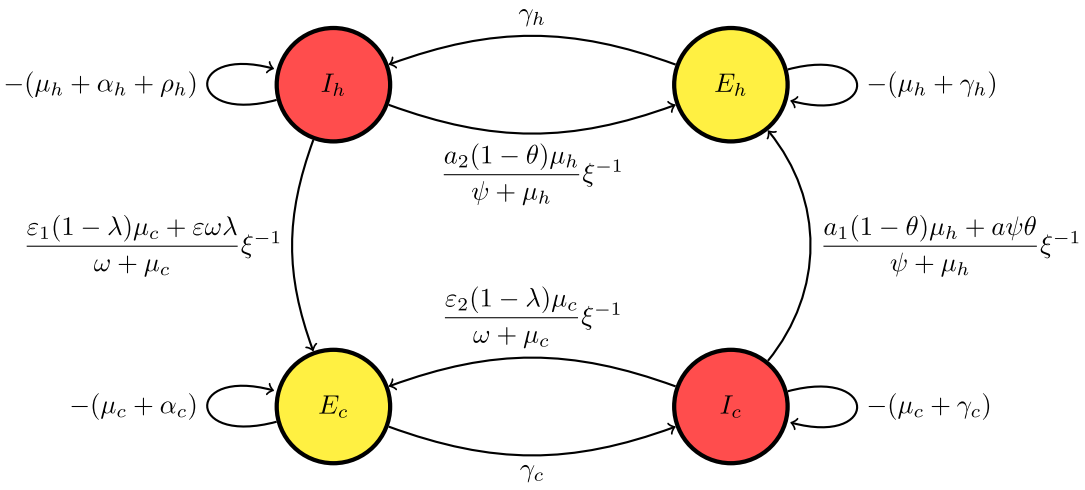


Fig. 3. (**A***) Digraph representation of $\mathcal{F}\xi^{-1}-\mathcal{M}$.

The approach outlined in de Camino-Beck et al. (2009), de Camino-Beck and Lewis (2008, 2007), Li and Schneider (2002) highlights that the $\mathbb{GT}$ method begins by separating new infections from other population dynamics. In this framework, $\mathcal{F}$ is an $n\times n$ matrix, while $\mathcal{M}$ is a non-singular $m\times m$ matrix. Both matrices are derived through partial derivatives and evaluated at the system's equilibrium. The specific forms of $\mathcal{F}$ and $\mathcal{M}$ are provided herein:

$$\mathcal{F}=\left[\frac{\partial\mathcal{F}_i}{\partial(E_h,I_h,E_c,I_c)}\right]_{E_0}\quad\text{and}\quad\mathcal{M}=\left[\frac{\partial\mathcal{M}_i}{\partial(E_h,I_h,E_c,I_c)}\right]_{E_0} \tag{6}$$

$$\mathcal{F}=\begin{bmatrix}0 & \frac{a_2(1-\theta)\mu_h}{\psi+\mu_h} & 0 & \frac{a_1(1-\theta)\mu_h+a\psi\theta}{\psi+\mu_h}\\ 0 & 0 & 0 & 0\\ 0 & \frac{\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda}{\omega+\mu_c} & 0 & \frac{\varepsilon_2(1-\lambda)\mu_c}{\omega+\mu_c}\\ 0 & 0 & 0 & 0\end{bmatrix}, \tag{7}$$

and

$$\mathcal{M}=\begin{bmatrix}(\mu_h+\gamma_h) & 0 & 0 & 0\\ -\gamma_h & (\mu_h+\alpha_h+\rho_h) & 0 & 0\\ 0 & 0 & (\mu_c+\gamma_c) & 0\\ 0 & 0 & -\gamma_c & (\mu_c+\alpha_c)\end{bmatrix}. \tag{8}$$

From which,

$$\mathcal{F}\xi^{-1}-\mathcal{M}=\begin{bmatrix}-(\mu_h+\gamma_h) & \frac{a_2(1-\theta)\mu_h}{\psi+\mu_h}\xi^{-1} & 0 & \frac{a_1(1-\theta)\mu_h+a\psi\theta}{\psi+\mu_h}\xi^{-1}\\ \gamma_h & -(\mu_h+\alpha_h+\rho_h) & 0 & 0\\ 0 & \frac{\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda}{\omega+\mu_c}\xi^{-1} & -(\mu_c+\gamma_c) & \frac{\varepsilon_2(1-\lambda)\mu_c}{\omega+\mu_c}\xi^{-1}\\ 0 & 0 & \gamma_c & -(\mu_c+\alpha_c)\end{bmatrix}, \tag{9}$$

2.3.2. Illustration of the matrix and digraph relationships in the model.

The matrix $\mathcal{F}\xi^{-1}-\mathcal{M}$ captures the transition dynamics between epidemiological compartments, with each element representing the rate of movement between states such as exposed, infected, and recovered individuals. The corresponding digraph visualizes these interactions, where nodes represent compartments, and directed edges denote transition rates. The nodes in the digraph, including E_h, I_h, E_c, and I_c, represent different disease states for human and cattle. Transition rates like γ_h and $\frac{\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda}{\omega+\mu_c}$ illustrate disease progression and transmission dynamics. Self-loops on nodes indicate internal transitions, such as natural death and recovery, while off-diagonal elements capture the movement between compartments, like from exposed to infectious states as illustrated in Fig. 3.

2.3.3. Digraph reduction and Gaussian elimination

Digraph reduction mimics Gaussian elimination on the matrix $\mathcal{F}\xi^{-1}-\mathcal{M}$ by simplifying the system through the elimination of intermediate nodes, analogous to removing off-diagonal matrix elements. A node becomes trivial when its self-loop is normalized to -1, typically reflecting constant rates like natural mortality. Two key rules guide the process: (1) normalize incoming edges to create a trivial node, and (2) replace paths through that node with direct connections, preserving the determinant, as clearly described in detail in the work of Oswald Sylas et al. (2025b). For example, node I_c is reduced by dividing its incoming edges by $(\mu_c+\gamma_c)$ and updating the connections accordingly. Fig. 4 illustrates these structural transformations within the digraph.

$$\text{where}\qquad \mathcal{V}^*=(\mu_c+\alpha_c)-\left\{\frac{\varepsilon_2(1-\lambda)\mu_c}{(\omega+\mu_c)(\mu_c+\gamma_c)}\right\}\gamma_c\xi^{-1} \tag{10}$$

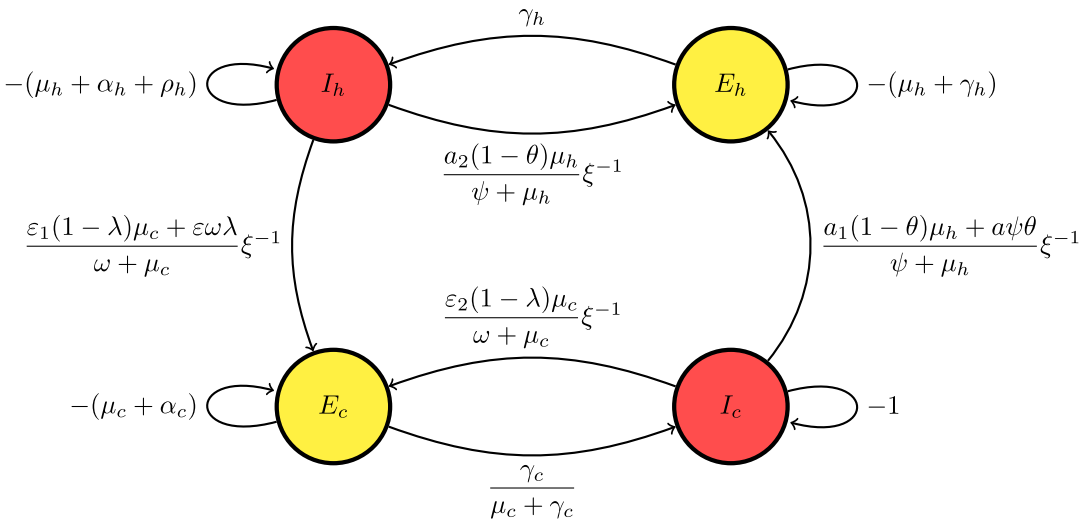


Fig. 4. (**A**) Creating a trivial node at node I_c.

By examining the connectivity and cyclic structure of these digraphs, in Figs. 4–7 particularly through Gaussian elimination and spectral radius analysis, the approach facilitates a more structured and transparent computation of $\mathcal{R}_e$. This not only simplifies the mathematical derivation but also enhances the interpretability of transmission dynamics by explicitly linking $\mathcal{R}_e$ to individual transmission routes within the model. For further methodological details and implementation, interested readers are referred to Oswald Sylas et al. (2025b,a).

$$-(\mu_h+\gamma_h)+\frac{\gamma_h}{(\mu_h+\alpha_h+\rho_h)}\left\{\frac{a_2(1-\theta)\mu_h}{\psi+\mu_h}\xi^{-1}+\frac{(a_1(1-\theta)\mu_h+a\psi\theta)(\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda)}{\mathcal{V}^*(\omega+\mu_c)(\psi+\mu_h)(\mu_c+\gamma_c)}\gamma_c\xi^{-2}\right\}=0 \tag{11}$$

From Eq. (10), denoting $\mathcal{J}_1=\mu_c+\alpha_c$, $\mathcal{J}_2=\left\{\frac{\varepsilon_2(1-\lambda)\mu_c}{(\omega+\mu_c)(\mu_c+\gamma_c)}\right\}\gamma_c$, such that $\mathcal{V}^*=\mathcal{J}_1-\mathcal{J}_2\xi^{-1}$, while from Eq. (11), $\mathcal{D}_1=\mu_h+\gamma_h$, $\mathcal{D}_2=\frac{\gamma_h}{\mu_h+\alpha_h+\rho_h}$, $\mathcal{C}=\frac{a_2(1-\theta)\mu_h}{\psi+\mu_h}$ and $\mathcal{B}=\frac{(a_1(1-\theta)\mu_h+a\psi\theta)(\varepsilon_1(1-\lambda)\mu_c+\varepsilon\omega\lambda)}{(\omega+\mu_c)(\psi+\mu_h)(\mu_c+\gamma_c)}\gamma_c$, now equation (11) becomes

$$-\mathcal{D}_1+\mathcal{D}_2\left\{\mathcal{C}\xi^{-1}+\frac{\mathcal{B}}{\mathcal{J}_1-\mathcal{J}_2\xi^{-1}}\xi^{-2}\right\}=0 \tag{12}$$

Simplifying equation (12), we get

$$-\mathcal{D}_1\mathcal{J}_1+\mathcal{D}_1\mathcal{J}_2\xi^{-1}+\mathcal{J}_1\mathcal{D}_2\mathcal{C}\xi^{-1}-\mathcal{J}_2\mathcal{D}_2\mathcal{C}\xi^{-2}+\mathcal{B}\mathcal{D}_2\xi^{-2}=0 \tag{13}$$

Fo simplification, one may multiply ξ^2 on equation (13) and observe that

$$-\mathcal{D}_1\mathcal{J}_1\xi^2+(\mathcal{D}_1\mathcal{J}_2+\mathcal{J}_1\mathcal{D}_2\mathcal{C})\xi+(-\mathcal{J}_2\mathcal{D}_2\mathcal{C}+\mathcal{B}\mathcal{D}_2)=0 \tag{14}$$

One may realize that from Eq. (14), the effective reproduction number of the model system (1), is given by

$$\mathcal{R}_e=\frac{1}{2\mathcal{D}_1\mathcal{J}_1}\left(\mathcal{D}_1\mathcal{J}_2+\mathcal{J}_1\mathcal{D}_2\mathcal{C}+\sqrt{(\mathcal{D}_1\mathcal{J}_2+\mathcal{J}_1\mathcal{D}_2\mathcal{C})^2+4\mathcal{D}_1\mathcal{J}_1(-\mathcal{J}_2\mathcal{D}_2\mathcal{C}+\mathcal{B}\mathcal{D}_2)}\right). \tag{15}$$

So, when there is no intervention strategy such that $\psi=0$, and $\omega=0$ which refer to the education campaign and vaccination rate, and without any treatment for individual human is implemented ($\rho_h=0$). Then, the basic reproduction number $\mathcal{R}_0$ will be

$$\mathcal{R}_0=\frac{1}{2\mathcal{D}_1\mathcal{J}_1}\left(\mathcal{D}_1\mathcal{J}_2+\mathcal{J}_1\mathcal{D}_2\mathcal{C}+\sqrt{(\mathcal{D}_1\mathcal{J}_2+\mathcal{J}_1\mathcal{D}_2\mathcal{C})^2+4\mathcal{D}_1\mathcal{J}_1(-\mathcal{J}_2\mathcal{D}_2\mathcal{C}+\mathcal{B}\mathcal{D}_2)}\right), \tag{16}$$

where, $\mathcal{B}\mathcal{D}_2=\frac{\gamma_c\gamma_h a_1}{\mu_c(\gamma_c+\mu_c)(\alpha_h+\mu_h)}$, $\mathcal{J}_2\mathcal{D}_2\mathcal{C}=\frac{\gamma_h a_2\varepsilon_2}{(\gamma_c+\mu_c)(\alpha_h+\mu_h)}$, $\mathcal{J}_1\mathcal{D}_2\mathcal{C}=\frac{(\alpha_c+\mu_c)\gamma_h a_2}{(\alpha_h+\mu_h)}$, and $\mathcal{D}_1\mathcal{J}_2=\frac{(\gamma_h+\mu_h)\varepsilon_2}{(\gamma_c+\mu_c)}$, $\mathcal{J}_1\mathcal{D}_1=(\alpha_c+\mu_c)(\gamma_h+\mu_h)$.

If $\mathcal{R}_0<1$, the implication is that, on average, an infectious individual induces less than one newly-infected individuals. Thus, with time the bTB disease could die out and the population would be freed from bTB. In other words, human and cattle population would remain free from the disease invasion if $\mathcal{R}_0<1$.

Conversely, if $\mathcal{R}_0>1$ then an average, there would be more than one newly-infected individuals induced by each infected individual. In this situation the infection continues to wreak havoc and the disease continues to spread in the wider population, which implies that the bTB disease will continue to spread in the population if $\mathcal{R}_0>1$.

2.4. Local stability of Disease-Free Equilibrium (DFE)

The local stability analysis of DFE enables us to grasp how a system behaves near the equilibrium point, but not necessarily at a given equilibrium point (Castillo-Chavez et al., 2002; Oswald Sylas, 2023; Sosoma et al., 2025; Irunde et al., 2023; Leandry et al., 2025b,a,c). In other words, local stability investigates the nature of the system surrounding the neighbourhood of the equilibrium point.

Theorem 2. *The bTB model system at DFE (E_0) is locally asymptotically stable if $\mathcal{R}_e<1$ and unstable if $\mathcal{R}_e>1$.*

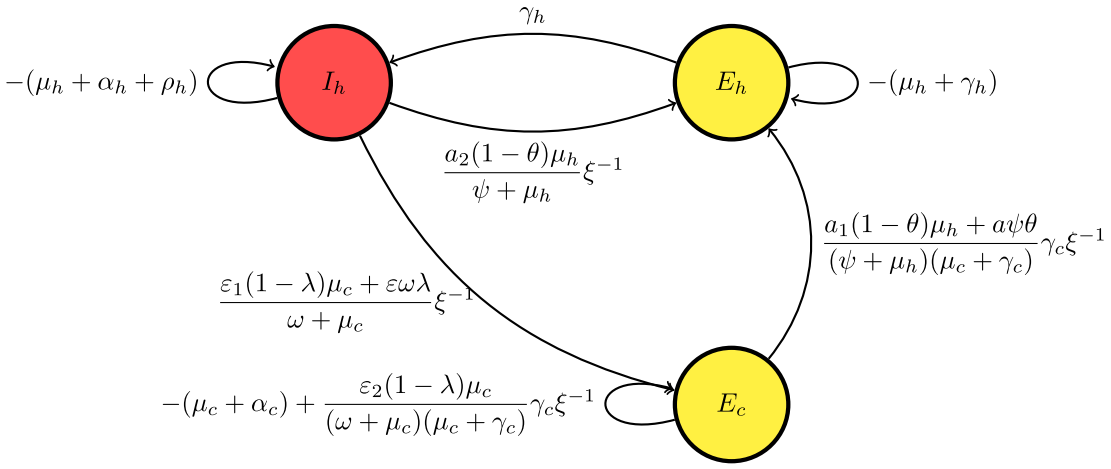


Fig. 5. (**B**) Digraph representation of $\mathcal{F}\xi^{-1}-\mathcal{M}$ without node I_c.

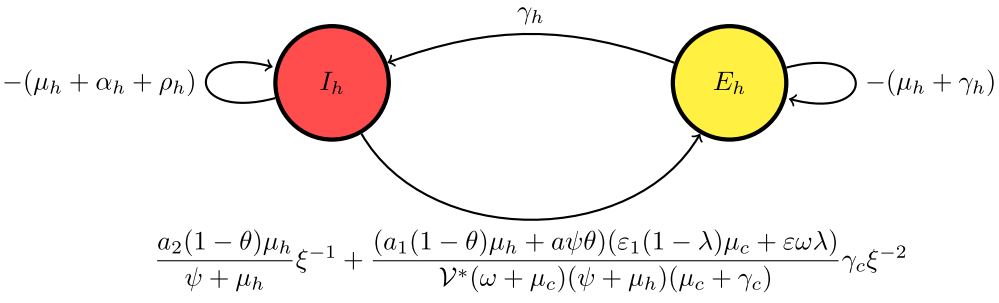


Fig. 6. (**C**) Digraph representation of $\mathcal{F}\xi^{-1}-\mathcal{M}$ without node E_c and I_c.

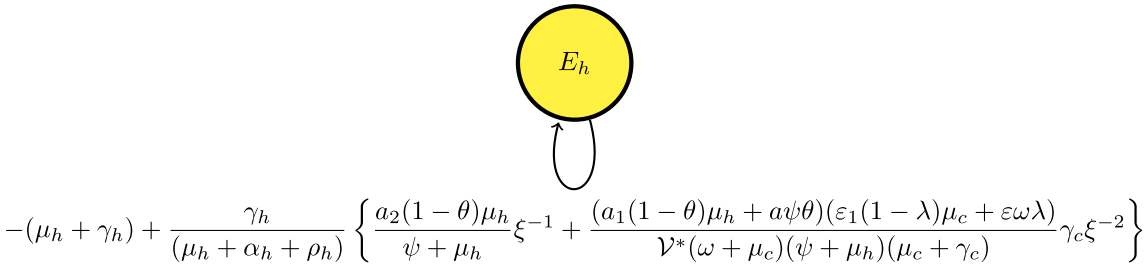


Fig. 7. (**D**) Digraph representation of $\mathcal{F}\xi^{-1}-\mathcal{M}$ without node I_h, E_c and I_c.

Proof. The Jacobian matrix of the model system (1) at Disease free equilibrium point (E_0) is given by;

$$J_{E_0}=\begin{pmatrix}-y_{11}&0&0&y_{14}&0&0&0&0&y_{19}\\ \psi&-\mu_h&0&0&0&0&0&0&y_{29}\\ 0&0&-y_{33}&y_{34}&0&0&0&0&y_{39}\\ 0&0&\gamma_h&-y_{44}&0&0&0&0&0\\ 0&0&0&\rho_h&-y_{55}&0&0&0&0\\ 0&0&0&y_{64}&0&-y_{66}&0&y_{68}&0\\ 0&0&0&y_{74}&0&\omega&-\mu_c&0&0\\ 0&0&0&y_{84}&0&0&0&-y_{88}&y_{89}\\ 0&0&0&0&0&0&0&\gamma_c&-y_{99},\end{pmatrix}$$

where, $y_{11}=\mu_h+\psi$, $y_{14}=\frac{-a_2(1-\theta)\mu_h}{\mu_h+\psi}$, $y_{19}=\frac{-a_1(1-\theta)\mu_h}{\mu_h+\psi}$, $y_{29}=\frac{-a\theta\psi}{\mu_h+\psi}$, $y_{33}=\mu_h+\gamma_h$,
$y_{34}=\frac{a_2(1-\theta)\mu_h}{\mu_h+\psi}$, $y_{39}=\frac{a_1(1-\theta)\mu_h}{\mu_h+\psi}$, $y_{44}=\mu_h+\alpha_h+\rho_h$, $y_{55}=\mu_h+\tau_h$, $y_{64}=\frac{-(1-\lambda)\varepsilon_1\mu_c}{\mu_c+\omega}$, $y_{66}=\mu_c+\omega$, $y_{68}=\frac{-(1-\lambda)\varepsilon_2\mu_c}{\mu_c+\omega}$, $y_{74}=\frac{-\lambda\varepsilon\omega}{\mu_c+\omega}$,
$y_{84}=\frac{(1-\lambda)\varepsilon_1\mu_c}{\mu_c+\omega}$, $y_{88}=(\mu_c+\gamma_c)$, $y_{89}=\frac{(1-\lambda)\varepsilon_2\mu_c}{\mu_c+\omega}$ and $y_{99}=\mu_c+\alpha_c$.

From the Jacobian matrix J_{E_0}, we observe that $-\mu_h, -(\mu_h+\tau_h)$, $-(\mu_h+\psi)$, $-(\mu_c+\omega)$ and $-\mu_c$ are the eigenvalues of the Jacobian J_{E_0}. Thus, excluding these columns and the corresponding rows, the remaining eigenvalues are computed. Then the reduced 4×4 matrix from J_{E_0} becomes:

$$Y=\begin{pmatrix} -y_{33} & y_{34} & 0 & y_{39} \\ \gamma_h & -y_{44} & 0 & 0 \\ 0 & y_{84} & -y_{88} & y_{89} \\ 0 & 0 & \gamma_c & -y_{99} \end{pmatrix}=\begin{pmatrix} G_1 & G_3 \\ G_4 & G_2 \end{pmatrix}.$$

Matrices G_1, G_2, G_3 and G_4 are defined as follows

$$G_1=\begin{pmatrix} -y_{33} & y_{34} \\ \gamma_h & -y_{44} \end{pmatrix}, G_2=\begin{pmatrix} -y_{88} & y_{89} \\ \gamma_c & -y_{99} \end{pmatrix}, G_3=\begin{pmatrix} 0 & y_{39} \\ 0 & 0 \end{pmatrix}, G_4=\begin{pmatrix} 0 & y_{84} \\ 0 & 0 \end{pmatrix}.$$

Apparently, that matrices G_3 and G_4 are singular matrices such that $|G_3|=0$ and $|G_4|=0$.

Now, it is enough to show that the disease free equilibrium point of the model system (1) is stable only if the trace and determinant of matrices G_1 and G_2 are negative and positive respectively. Trace and determinant of matrix G_1 denoted by $\mathrm{Tr}(G_1)$ and $\det(G_1)$ are respectively given by:

$$\mathrm{Tr}(G_1)=-(y_{33}+y_{44})=-(2\mu_h+\gamma_h+\alpha_h+\rho_h)<0.$$

$$\det(G_1)=y_{33}y_{44}-\gamma_h y_{34}=(\mu_h+\gamma_h)(\mu_h+\alpha_h+\rho_h)-\frac{a_2(1-\theta)\mu_h\gamma_h}{\mu_h+\psi}$$

such that $\dfrac{a_2(1-\theta)\mu_h\gamma_h}{\mu_h+\psi}=A(\gamma_h+\mu_h)(\alpha_h+\mu_h+\rho_h)$,

where,

$$A=\mathcal{R}_e-\frac{\gamma_c\gamma_h\mu_c\mu_h\varepsilon_1 a_1(a\psi\theta+1-\theta)(\varepsilon\omega\lambda+1-\lambda)}{(\gamma_h+\mu_h)(\alpha_h+\mu_h+\rho_h)(\psi+\mu_h)\left[((\omega+\mu_c)(\alpha_c+\mu_c)^2\mathcal{R}_e-\gamma_c\mu_c\varepsilon_2(1-\lambda))\right]}.$$

Trace and determinant of matrix G_2 denoted by $\mathrm{Tr}(G_2)$ and $\det(G_2)$ are respectively given by:

$$\mathrm{Tr}(G_2)=-(y_{88}+y_{99})=-(2\mu_c+\gamma_c+\alpha_c)<0.$$

$$\det(G_2)=y_{88}y_{99}-\gamma_c y_{89}=(\mu_c+\gamma_c)(\mu_c+\alpha_c)-\frac{\varepsilon_2(1-\lambda)\mu_c\gamma_c}{\mu_c+\omega},$$

such that $\dfrac{\varepsilon_2(1-\lambda)\mu_c\gamma_c}{\mu_c+\omega}=B(\alpha_c+\mu_c)^2$,

where, $B=\mathcal{R}_e-\dfrac{\gamma_c\gamma_h\mu_c\mu_h\varepsilon_1 a_1(a\psi\theta+1-\theta)(\varepsilon\omega\lambda+1-\lambda)}{(\omega+\mu_c)(\alpha_c+\mu_c)^2\left[(\gamma_h+\mu_h)(\alpha_h+\mu_h+\rho_h)(\psi+\mu_h)\mathcal{R}_e-\gamma_h\mu_h a_2(1-\theta)\right]}$.

Thus $\det(G_1)>0$ and $\det(G_2)>0$ if and only if $\mathcal{R}_e<1$. Since the traces of matrix G_1 and G_2 are negative and the determinants are strictly greater than zero when $\mathcal{R}_e<1$, then the disease-free equilibrium point E_0 is locally asymptotically stable when $\mathcal{R}_e<1$ and unstable otherwise.

Epidemiologically, it implies that the bTB disease can be eliminated in the endemic area when $\mathcal{R}_e<1$ particularly when vaccination rate (ω) is kept constant without limitation and the education campaign rate (ψ) is also executed for every individual. Conversely, if $\mathcal{R}_e>1$ then each infectious individual produces more than one new infected individual, implying that, the disease can spread rapidly in the whole population. □

2.5. Global stability of Disease-Free Equilibrium (DFE)

This subsection analyses the global stability pertaining to DFE aimed to establish the asymptotic behaviour of the model system (1) beyond just the neighbourhood points of the model disease-free steady state.

Theorem 3. *The disease-free equilibrium point is g.a.s if $\mathcal{R}_e<1$ and unstable if $\mathcal{R}_e>1$.*

Proof. The approach of Castillo-Chavez and Song (2004), Leandry and Mureithi (2023), Castillo-Chavez et al. (2002), Ndendya Jufren Zakayo et al. (2025f) is used to analyse the global stability of DFE of the model system (1). Using this approach the model system (1) is written as follows

$$\begin{cases} \dfrac{d\zeta_n}{dt}=M_1(\zeta_n-\zeta_{E_0})+M_2\zeta_i, \\ \dfrac{d\zeta_i}{dt}=M_3\zeta_i. \end{cases} \tag{17}$$

where ζ_n stands for classes that do not transmit bTB disease, that is $\zeta_n=(S_{nh},S_{eh},R_h,S_{nv},S_v)^T$ and ζ_i stands for classes that can transmit bTB disease, that is $\zeta_i=(I_h,E_h,E_c,I_c)^T$. Here T stands for the transposition of ζ_n and ζ_i, also ζ_{E0} is ζ_n at DFE (E_0). The matrix M_1 is obtained from differentiating non-transmitting equations of the model system (1) with respect to non-transmitting variables at E_0 whereas the matrix M_2 is obtained by differentiating the non transmitting equations of the model system (1) with respect to the transmitting variables.

The disease-free equilibrium point (E_0) is globally asymptotically stable if the eigenvalues of M_1 are real negative and M_3 is a Metzler stable matrix and, its off-diagonal elements are non-negative. So, from the model system (1) yields the following:

$$\zeta_n-\zeta_{E_0}=\begin{pmatrix} S_{nh} \\ S_{eh} \\ R_h \\ S_{nv} \\ S_v \end{pmatrix}-\begin{pmatrix} \dfrac{\pi_h}{\mu_h+\psi} \\ \dfrac{\psi}{\mu_h}\left(\dfrac{\pi_h}{\psi+\mu_h}\right) \\ 0 \\ \dfrac{\pi_c}{\mu_c+\omega} \\ \dfrac{\omega}{\mu_c}\left(\dfrac{\pi_c}{\omega+\mu_c}\right) \end{pmatrix}=\begin{pmatrix} S_{nh}-\dfrac{\pi_h}{\mu_h+\psi} \\ S_{eh}-\dfrac{\psi}{\mu_h}\left(\dfrac{\pi_h}{\psi+\mu_h}\right) \\ R_h \\ S_{nv}-\dfrac{\pi_c}{\mu_c+\omega} \\ S_v-\dfrac{\omega}{\mu_c}\left(\dfrac{\pi_c}{\omega+\mu_c}\right) \end{pmatrix}. \tag{18}$$

Also,

$$M_1 = \frac{\partial \zeta_n(E_0)}{\partial(S_{nh}, S_{eh}, R_h, S_{nv}, S_v)} = \begin{pmatrix} -(\mu_h+\psi) & 0 & 0 & 0 & 0 \\ \psi & -\mu_h & 0 & 0 & 0 \\ 0 & 0 & -(\mu_h+\tau_h) & 0 & 0 \\ 0 & 0 & 0 & -(\mu_c+\omega) & 0 \\ 0 & 0 & 0 & \omega & -\mu_c \end{pmatrix},$$

$$M_2 = \frac{\partial \zeta_n}{\partial(E_h, I_h, E_c, I_c)} = \begin{pmatrix} 0 & -(1-\theta)a_2\frac{S_{nh}}{N_h} & 0 & -(1-\theta)a_1\frac{S_{nh}}{N_h} \\ 0 & 0 & 0 & \theta a\frac{S_{nh}}{N_h} \\ 0 & -(1-\lambda)\varepsilon_2\frac{S_{nv}}{N_c} & 0 & -(1-\lambda)\varepsilon_1\frac{S_{nv}}{N_c} \\ 0 & \lambda\varepsilon\frac{S_v}{N_c} & 0 & 0 \end{pmatrix},$$

Since the matrix M_1 is a lower triangular matrix, then the eigenvalues will be the diagonal entries presented as follows: $\xi_1 = -(\mu_h+\psi), \xi_2 = -\mu_h, \xi_3 = -(\mu_c+\omega)$ and $\xi_4 = -\mu_c$ which are all negative and real. This shows that at the DFE the system $\frac{d\zeta_n}{dt} = M_1(\zeta_n - \zeta_{E_0}) + M_2\zeta_i$ is globally asymptotically stable. Furthermore, The matrix M_3 is obtained by differentiating the transmitting equations of the model system (1) with respect to the transmitting variables which gives

$$M_3 = \begin{pmatrix} -(\mu_h+\gamma_h) & 0 & 0 & a\theta\frac{S_{eh}}{N_h} + a_1(1-\theta)\frac{S_{nh}}{N_h} \\ \gamma_h & -(\mu_h+\alpha_h+\rho_h) & 0 & 0 \\ 0 & \varepsilon\lambda\frac{S_v}{N_c} + \varepsilon_1(1-\lambda)\frac{S_{nv}}{N_c} & -(\mu_c+\gamma_c) & \varepsilon_2(1-\lambda)\frac{S_{nv}}{N_c} \\ 0 & 0 & \gamma_c & -(\mu_c+\alpha_c) \end{pmatrix}.$$

Testing whether the matrix M_3 is a Metzler stable matrix requires applying the approach as propounded by Kamgang and Sallet (2005), Dumont et al. (2008) by using he following Lemma:

Lemma 2.3. *Let G be a square Metzler matrix stable which can be written in block form*

$$G = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A and D are square matrices. Then, G is a Metzler stable matrix if and only if matrix A and $D - CA^{-1}B$ or matrix D and $A - BD^{-1}C$ are Metzler stable.

By comparing the matrix M_3 and a square Metzler matrix G, then the matrices A, B, C and D are expressed as follows:

$$A = \begin{pmatrix} -(\mu_h+\gamma_h) & 0 \\ \gamma_h & -(\mu_h+\alpha_h+\rho_h) \end{pmatrix}, D = \begin{pmatrix} -(\mu_c+\gamma_h) & \varepsilon_2(1-\lambda)\frac{S_{nv}}{N_c} \\ \gamma_c & -(\mu_c+\alpha_c) \end{pmatrix},$$

$$B = \begin{pmatrix} 0 & a\theta\frac{S_{eh}}{N_h} + a_1(1-\theta)\frac{S_{nh}}{N_h} \\ 0 & 0 \end{pmatrix}, C = \begin{pmatrix} 0 & \varepsilon\lambda\frac{S_v}{N_c} + \varepsilon_1(1-\lambda)\frac{S_{nv}}{N_c} \\ 0 & 0 \end{pmatrix}.$$

After some computations and simplifications,

$$A - BD^{-1}C = \begin{pmatrix} -(\mu_h+\gamma_h) & 0 \\ \gamma_h & -(\mu_h+\alpha_h+\rho_h) \end{pmatrix}.$$

The matrix $A - BD^{-1}C$ is said to be Metzler stable if and only if $(\mu_h+\gamma_h)(\mu_h+\alpha_h+\rho_h) \geq 0$. The matrix M_1 has all eigenvalues real and negative, and M_3 is a Metzler matrix since its off-diagonal elements are non-negative, implying that $|A - BD^{-1}C| \geq 0 \Rightarrow M_3(i,j) \geq 0$ for all $i \neq j$. Therefore, the disease-free equilibrium is globally asymptotically stable when $\mathcal{R}_e < 1$; otherwise, it is unstable.

Biologically, this indicates that bovine tuberculosis (bTB) will eventually be eliminated from the population if $\mathcal{R}_e < 1$, regardless of the initial infection level. However, if $\mathcal{R}_e > 1$, the disease will persist and continue to spread due to ongoing new infections. □

2.6. Endemic Equilibrium Point (EEP)

The endemic equilibrium (EE), denoted by E_*, is defined as $E_* = \left(S_{nh}^*,\ S_{eh}^*,\ E_h^*,\ I_h^*,\ R_h^*,\ S_{nv}^*,\ S_v^*,\ E_c^*,\ I_c^*\right)$, which represents a state where the infection persists within the population. At this equilibrium, the disease remains endemic and cannot be eradicated. Specifically, for the case of bovine tuberculosis (bTB), the number of infectious individuals is nonzero, i.e., $E_h \neq 0,\ I_h \neq 0,\ R_h \neq 0,\ E_c \neq 0,\ I_c \neq 0$. By setting each equation

in the model system (1) to zero, the steady state of the system can be determined and shown as follows

$$\begin{aligned}
S_{nh}^* &= \frac{\pi_h}{\mu_h+\psi+\varphi_1^*},\\
S_{eh}^* &= \frac{\psi\pi_h}{(\mu_h+\psi+\varphi_1^*)(\mu_h+\varphi_2^*)},\\
E_h^* &= \frac{(\psi\pi_h\varphi_1^*\varphi_2^*)(\mu_h+\alpha_h+\rho_h)(\mu_h+\tau_h)+\psi\varphi_2^*}{(\psi+\mu_h+\varphi_1^*)(\mu_h+\alpha_h+\rho_h)(\mu_h+\tau_h)(\mu_h+\varphi_2^*)+\gamma_h\rho_h\tau_h(\mu_h+\varphi_2^*)},\\
I_h^* &= \frac{\pi_h\gamma_h(\mu_h+\tau_h)(\mu_h\varphi_1^*+\varphi_1^*\varphi_2^*+\psi\varphi_2^*)}{(\psi+\mu_h+\varphi_1^*)(\mu_h+\alpha_h+\rho_h)(\mu_h+\tau_h)(\mu_h+\varphi_2^*)+\gamma_h\rho_h\tau_h(\mu_h+\varphi_2^*)},\\
R_h^* &= \frac{\pi_h\gamma_h\rho_h(\mu_h\varphi_1^*+\varphi_2^*\varphi_1^*+\psi\varphi_2^*)}{(\psi+\mu_h+\varphi_1^*)(\mu_h+\alpha_h+\rho_h)(\mu_h+\tau_h)(\mu_h+\varphi_2^*)+\gamma_h\rho_h\tau_h(\mu_h+\varphi_2^*)},\\
S_{nv}^* &= \frac{\pi_c\omega}{\mu_c+\omega+\sigma_1^*},\\
S_v^* &= \frac{\omega\pi_c}{(\mu_c+\omega+\sigma_1^*)(\mu_c+\sigma_2^*)},\\
E_c^* &= \frac{\pi_c(\mu_c\sigma_1^*+\omega\sigma_2^*+\sigma_1^*\sigma_2^*)}{(\mu_c+\omega+\sigma_1^*)(\mu_c+\gamma_c)(\mu_c+\sigma_2^*)},\\
I_c^* &= \frac{\gamma_h\pi_c(\mu_c\sigma_1^*+\omega\sigma_2^*+\sigma_1^*\sigma_2^*)}{(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\omega+\sigma_1^*)(\mu_c+\sigma_2^*)}.
\end{aligned} \tag{19}$$

where , $\varphi_1^* = (1-\theta)\left(a_1\frac{I_c^*}{N_h^*}+a_2\frac{I_h^*}{N_h^*}\right)$, $\varphi_2^* = a\theta\frac{I_c^*}{N_h^*}$, $\sigma_1^* = (1-\lambda)\left(\varepsilon_1\frac{I_h^*}{N_c^*}+\varepsilon_2\frac{I_c^*}{N_c^*}\right)$ and $\sigma_2^* = \varepsilon\lambda\frac{I_h^*}{N_c^*}$.

From Eq. (19), the value of I_h^* and I_c^* are substituted into $\sigma_1^* - (1-\lambda)\left(\varepsilon_1\frac{I_h^*}{N_c^*}+\varepsilon_2\frac{I_c^*}{N_c^*}\right) = 0$, and after some algebraic computations and simplifications, we get $A\sigma_1^{*3}+B\sigma_1^{*2}+C\sigma_1^* = 0$. Upon algebraic manipulations, $A\sigma_1^{*3}+B\sigma_1^{*2}+C\sigma_1^* = 0$ which results to $\sigma_1^*(A\sigma_1^{*2}+B\sigma_1^*+C) = 0$, where $\sigma_1^* = 0$ or $A\sigma_1^{*2}+B\sigma_1^*+C = 0$, such that,

$$A = \mu_c + \frac{a\theta\gamma_h\pi_c}{N_h(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)}(\mu_c+\sigma_2^*),$$

$$\begin{aligned}
B = &-\frac{a\theta\gamma_h^2\pi_c^2(\mu_c^2+2\mu_c\sigma_2^*+\sigma_2^{*^2})}{N_h[(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)]^2}+\omega\sigma_2^*+(\mu_c+\omega)(\mu_c+\mu_h)\\
&+\frac{\gamma_h\pi_c(-N_h\mu_c^2-\mu_c\sigma_2^*+2a\theta\mu_c(\omega+\mu_c)+a\theta\omega\sigma_2^*)}{N_h(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)}\\
&-\frac{2a\theta(1-\lambda)\gamma_h^2\pi_h\pi_c\varepsilon_1\mu_c(\psi+\varphi_1^*)(\mu_h+\tau_h)}{N_hN_c(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)(\rho_h\tau_h\gamma_h+(\rho_h+\alpha_h+\mu_h)(\psi+\mu_h+\varphi_1^*)(\mu_h+\tau_h))}\\
&-\frac{(1-\lambda)\gamma_h\pi_h\varepsilon_1\mu_h(\mu_h+\tau_h)\varphi_1^*}{N_c(\rho_h\tau_h\gamma_h+(\rho_h+\alpha_h+\mu_h)(\psi+\mu_h+\varphi_1^*)(\mu_h+\tau_h))},
\end{aligned}$$

and

$$\begin{aligned}
C = &-\frac{\gamma_h\pi_c^2(\mu_c\mu_h(\omega+\mu_c)+\mu_h(\omega+\mu_c)\sigma_2^*+\omega\sigma_2^{*^2}+\omega\sigma_2^*\mu_c)}{(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)}\\
&-\frac{\gamma_h\pi_c(N_h\mu_c^2+\mu_c\sigma_2^*+2a\theta\mu_c(\omega+\mu_c)+a\theta\omega\sigma_2^*)}{N_h(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)}\\
&-\frac{a\theta(1-\lambda)\gamma_h^2\pi_h\pi_c\varepsilon_1\mu_c(\psi+\varphi_1^*)(\mu_h+\tau_h)}{N_hN_c(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)(\rho_h\tau_h\gamma_h+(\rho_h+\alpha_h+\mu_h)(\psi+\mu_h+\varphi_1^*)(\mu_h+\tau_h))}\\
&-\frac{2(1-\lambda)\gamma_h\pi_h\varepsilon_1\mu_h(\mu_h+\tau_h)(\omega+\mu_c)\varphi_1^*(1+2\omega-2\mu_c)}{N_c(\rho_h\tau_h\gamma_h+(\rho_h+\alpha_h+\mu_h)(\psi+\mu_h+\varphi_1^*)(\mu_h+\tau_h))}\\
&-\frac{2a\theta\gamma_h^2\pi_c^2\omega\sigma_2^*(\mu_c^2+\mu_c\sigma_2^*+\sigma_2^{*^2})}{N_h[(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)]^2}\\
&+\left(\frac{\omega\sigma_2^*}{\mu_c+\mu_h}+\mu_h\right)\frac{(\mu_c+\omega)^2\gamma_h\pi_ca\theta\omega\sigma_2^*(\rho_h+\alpha_h+\mu_h)(\psi+\mu_h)(\mu_h+\tau_h)}{(\mu_c^2+2\mu_c\sigma_2^*+\sigma_2^{*^2})N_h(\mu_c+\alpha_c)(\gamma_c+\mu_c)(\mu_c+\sigma_2^*)}(1-\mathcal{R}_e)\mathcal{R}_e
\end{aligned} \tag{20}$$

For $\sigma_1^* = (1-\lambda)\left(\varepsilon_1\frac{I_h^*}{N_c^*}+\varepsilon_2\frac{I_c^*}{N_c^*}\right) = 0$, corresponds to DFE, whereas $A\sigma_1^{*2}+B\sigma_1^*+C = 0$ can also be written in the form:

$$\sigma_1^* = \frac{-B\pm\sqrt{B^2-4AC}}{2},$$

which satisfies Endemic Equilibrium. The value of A is strictly positive. Depending on the signs of B and C, there are three cases to consider as having positive real root of the force of infection (σ_1^*) as follows:

Case 1: If $B < 0$ then model system (1) has a stable endemic equilibrium point when $C < 0$. This equilibrium happens when $\mathcal{R}_e > 1$ as interpreted from (20). In this case backward bifurcation is not possible due to the absence of multiple equilibria.

Case 2: Exactly one endemic equilibrium point.

Suppose $B < 0$ and $C = 0$ or $B^2 - 4AC = 0$. In other words, the polynomial $A\sigma_1^{*2} + B\sigma_1^* + C = 0$ has just one positive root and hence the model system (1) has unique endemic equilibrium point.

Case 3: Two endemic equilibria.

If $B < 0$, $C > 0$ and $B^2 - 4AC > 0$, then the polynomial $A\sigma_1^{*2} + B\sigma_1^* + C = 0$ has two positive real roots. In other words, the model system (1) has two endemic equilibria and hence there is a possibility of backward bifurcation. These three cases are summarized under the following theorem:

Theorem 4. *The number of positive endemic equilibria of bovine tuberculosis model (1) is hereunder summarized thus:*

(i) If $C < 0$, $\mathcal{R}_e > 1$, then the system has a unique endemic equilibrium.
(ii) If $B < 0$ and $C = 0$ or $B^2 - 4AC = 0$, then the system has exactly one endemic equilibrium.
(iii) If $B < 0$, $C > 0$ and $B^2 - 4AC > 0$, then the system has exactly two endemic equilibria.
(iv) Otherwise there are no endemic equilibria, i.e. when $AC > 0$ and $B > 0$.

2.7. Stability of Endemic Equilibrium Point (EEP) of model with interventions

In the epidemiological models, forward or backward bifurcation has a vital implication for biological control measures of infectious diseases. The bifurcation analysis of the equilibrium points reveals whether the disease is completely reducible or persistent in the afflicted population. The bifurcation analysis occurs at disease-free equilibrium using Centre Manifold theory as presented in Castillo-Chavez et al. (2002). This is done by re-naming the state variables of the model system (1) as follows:

Let $x_1 = S_{nh}$, $x_2 = S_{eh}$, $x_3 = E_h$, $x_4 = I_h$, $x_5 = R_h$, $x_6 = S_{nv}$, $x_7 = S_v$, $x_8 = E_c$ and $x_9 = I_c$ and the resulting system can be written in the form of

$$\frac{dX_i}{dt} = F(X_i),$$

where, $X_i = (x_1, x_2, \ldots, x_9)^T$, $F = (f_1, f_2, \ldots, f_9)^T$ and $(.)^T$ denotes a matrix transpose. The model system (1) takes the form:

$$\begin{cases}
\frac{dx_1}{dt} = f_1 = & \pi_h - (\mu_h + \psi)x_1 - (1-\theta)\left(a_1\frac{x_9x_1}{N_h} + a_2\frac{x_4x_1}{N_h}\right), \\
\frac{dx_2}{dt} = f_2 = & \psi x_1 - \mu_h x_2 - a\theta\frac{x_9x_2}{N_h}, \\
\frac{dx_3}{dt} = f_3 = & (1-\theta)\left(a_1\frac{x_9x_1}{N_h} + a_2\frac{x_4x_1}{N_h}\right) + a\theta\frac{x_9x_2}{N_h} - (\mu_h + \gamma_h)x_3 + \tau_h x_5, \\
\frac{dx_4}{dt} = f_4 = & \gamma_h x_3 - (\mu_h + \alpha_h + \rho_h)x_4, \\
\frac{dx_5}{dt} = f_5 = & \rho_h x_4 - (\mu_h + \tau_h)x_5, \\
\frac{dx_6}{dt} = f_6 = & \pi_c - (\mu_c + \omega)x_6 - (1-\lambda)\left(\varepsilon_1\frac{x_4x_6}{N_c} + \varepsilon_2\frac{x_9x_6}{N_c}\right), \\
\frac{dx_7}{dt} = f_7 = & \omega x_6 - \mu_c x_7 - \varepsilon\lambda\frac{x_4x_7}{N_c}, \\
\frac{dx_8}{dt} = f_8 = & (1-\lambda)\left(\varepsilon_1\frac{x_4x_6}{N_c} + \varepsilon_2\frac{x_9x_6}{N_c}\right) + \varepsilon\lambda\frac{x_4x_7}{N_c} - (\mu_c + \gamma_c)x_8, \\
\frac{dx_9}{dt} = f_9 = & \gamma_c x_8 - (\mu_c + \alpha_c)x_9,
\end{cases} \tag{21}$$

where $N_h = x_1 + x_2 + x_3 + x_4 + x_5$ and $N_c = x_6 + x_7 + x_8 + x_9$.

Let ε_2^* be the bifurcation parameter, then, the system (21) is a system of model equations at disease free equilibrium point when $\varepsilon_2 = \varepsilon_2^*$ with $\mathcal{R}_0 = 1$ in Eq. (16). Then, the system (21) is transformed with $\varepsilon_2 = \varepsilon_2^*$ that has a simple zero eigenvalue, and Centre Manifold Theory is served to analyse the dynamics of (21) near $\varepsilon_2 = \varepsilon_2^*$. Thus, the Jacobian matrix of the system (21) at disease-free equilibrium denoted by $J(\varepsilon_2^*)$ is given by

$$J(\varepsilon_2^*) = \begin{pmatrix}
-R_{11} & 0 & 0 & -R_{14} & 0 & 0 & 0 & 0 & -R_{19} \\
\psi & -\mu_h & 0 & 0 & 0 & 0 & 0 & 0 & -R_{29} \\
0 & 0 & -R_{33} & R_{34} & 0 & 0 & 0 & 0 & R_{39} \\
0 & 0 & \gamma_h & -R_{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \rho_h & -R_{55} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -R_{64} & 0 & -R_{66} & 0 & -R_{68} & 0 \\
0 & 0 & 0 & -R_{74} & 0 & \omega & -\mu_c & 0 & 0 \\
0 & 0 & 0 & R_{84} & 0 & 0 & 0 & -R_{88} & R_{89} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \gamma_c & -R_{99}
\end{pmatrix},$$

where, $R_{11} = \mu_h + \psi$, $R_{14} = R_{34} = \frac{a_2(1-\theta)\mu_h}{\mu_h + \psi}$, $R_{19} = R_{39} = \frac{a_1(1-\theta)\mu_h}{\mu_h + \psi}$, $R_{29} = \frac{a\theta\psi}{\mu_h + \psi}$, $R_{33} = \mu_h + \gamma_h$, $R_{64} = R_{84} = \frac{(1-\lambda)\varepsilon_1\mu_c}{\mu_c + \omega}$, $R_{66} = \mu_c + \omega$, $R_{74} = \frac{\lambda\varepsilon\omega}{\mu_c + \omega}$, $R_{88} = (\mu_c + \gamma_c)$, $R_{68} = R_{89} = \frac{(1-\lambda)\varepsilon_2\mu_c}{\mu_c + \omega}$ and $R_{99} = \mu_c + \alpha_c$.

Now, the right and left eigenvectors, which is associated with the zero eigenvalues calculated. The right eigenvector associated with the zero eigenvalue is given by $w = (w_1, w_2, \dots, w_9)^T$, which results in the following:

$$\begin{cases} -R_{11}w_1 - R_{14}w_4 - R_{19}w_9 = 0, \\ \psi w_1 - \mu_h w_2 - R_{29}w_9 = 0, \\ -R_{33}w_3 + R_{34}w_4 + R_{39}w_9 = 0, \\ \gamma_h w_3 - R_{44}w_4 = 0, \\ \rho_h w_4 - R_{55}w_5 = 0, \\ -R_{64}w_4 - R_{66}w_6 - R_{68}w_8 = 0, \\ -R_{74}w_4 + \omega w_6 - \mu_c w_7 = 0, \\ R_{84}w_4 - R_{88}w_8 + R_{89}w_9 = 0, \\ \gamma_c w_8 - R_{99}w_9 = 0. \end{cases} \tag{22}$$

Solving the system (22) for $w_{i's}, i = 1, 2, \dots, 9$, gives the following right eigenvectors

$$\begin{aligned} w_1 &= \frac{-1}{R_{11}}(R_{14}w_4 + R_{19}w_9) \implies w_1 < 0, \\ w_2 &= \frac{-1}{\mu_h R_{11}}\left[(R_{11}R_{29} + \psi R_{19})w_9 + \psi R_{14}w_4\right] \implies w_2 < 0, \\ w_3 &= \frac{1}{R_{33}}(R_{34}w_4 + R_{39}w_9) \implies w_3 > 0, \\ w_4 &= \frac{\gamma_h w_3}{R_{44}} \implies w_4 > 0, \\ w_5 &= \frac{\rho_h \gamma_h}{R_{33}R_{44}R_{55}}(R_{34}w_4 + R_{39}w_9) \implies w_5 > 0, \\ w_6 &= \frac{-1}{R_{66}}\left[\frac{R_{64}\gamma_h}{R_{33}R_{44}}(R_{34}w_4 + R_{39}w_9) + \frac{R_{68}}{R_{88}}(R_{84}w_4 + R_{89}w_9)\right] \implies w_6 < 0, \\ w_7 &= \frac{-R_{74}w_4}{\mu_c} - \omega\left[\frac{R_{64}\gamma_h(R_{34}w_4 + R_{39}w_9)}{\mu_c R_{33}R_{44}R_{66}} + \frac{R_{68}(R_{84}\omega_4 + R_{89}w_9)}{\mu_c R_{66}R_{88}}\right] \implies w_7 < 0, \\ w_8 &= \frac{1}{R_{88}}(R_{84}w_4 + R_{89}w_9) \implies w_8 > 0, \\ w_9 &= \frac{\gamma_c}{R_{88}R_{99}}(R_{84}w_4 + R_{89}w_9) \implies w_9 > 0. \end{aligned}$$

Furthermore, to calculate the left eigenvectors given by $v = [v_1, v_2, \dots, v_9]^T$ which satisfy $v.w = 1$, the matrix $J(\varepsilon_2^*)$ is transposed and become

$$J(\varepsilon_2^*)^T = \begin{pmatrix} -R_{11} & \psi & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\mu_h & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -R_{33} & \gamma_h & 0 & 0 & 0 & 0 & 0 \\ R_{14} & 0 & R_{34} & -R_{44} & \rho_h & R_{64} & R_{74} & R_{84} & 0 \\ 0 & 0 & 0 & 0 & -R_{55} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -R_{66} & \omega & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\mu_c & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & R_{68} & 0 & -R_{88} & \gamma_c \\ -R_{19} & -R_{29} & R_{39} & 0 & 0 & 0 & 0 & R_{89} & -R_{99} \end{pmatrix},$$

Then solving $J(\varepsilon_2^*)^T$, the following are obtained

$$\begin{cases} -R_{11}v_1 + \psi v_2 = 0, \\ -\mu_h v_2 = 0, \\ -R_{33}v_3 + \gamma_h v_4 = 0, \\ R_{14}v_1 + R_{34}v_3 - R_{44}v_4 + \rho_h v_5 + R_{64}v_6 + R_{74}v_7 + R_{84}v_8 = 0, \\ -R_{55}v_5 = 0, \\ -R_{66}v_6 + \omega v_7 = 0, \\ -\mu_c v_7 = 0, \\ R_{68}v_6 - R_{88}v_8 + \gamma_c v_9 = 0, \\ R_{19}v_1 + R_{29}v_2 + R_{39}v_3 + R_{89}v_8 + R_{99}v_9 = 0. \end{cases} \tag{23}$$

Solving the system (22) for $v_{i's}, i = 1, 2, \ldots, 9$, yields the following left eigenvectors

$$\begin{cases} v_1 &= 0, \\ v_2 &= 0, \\ v_3 &= \dfrac{\gamma_h}{R_{33}} v_4 \\ v_4, &= \dfrac{R_{34}}{R_{44}} v_3 + \dfrac{R_{84}\gamma_c}{R_{44}R_{88}} v_9, \\ v_5 &= 0, \\ v_6 &= 0, \\ v_7 &= 0, \\ v_8 &= \dfrac{\gamma_c}{R_{88}} v_9, \\ v_9 = v_9 &> 0. \end{cases} \tag{24}$$

From Castillo-Chavez et al. (2002), Theorem 4:1 is used to establish the conditions for the existence of forward or backward bifurcations of the endemic equilibrium point near $\mathcal{R}_0 = 1$:

Lemma 2.4. *Consider the following general system of ordinary differential equations with a parameter β such that*

$$\frac{dx}{dt} = f(x, \beta), \quad f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \text{ and } f \in \mathbb{C}^2(\mathbb{R}^n \times \mathbb{R}), \tag{25}$$

such that $f(0, \beta) \equiv 0$, where 0 is an equilibrium point of the system with the following conditions:

1. $A = D_x f(0,0) = \left(\dfrac{\partial f_i}{\partial x_i}(0,0)\right)$ *is the linearization matrix of the model system (1) around the equilibrium 0 with β evaluated at 0.*
2. *Zero is a simple eigenvalue of A, and all other eigenvalues of A have negative real parts.*
3. *Matrix A has a non-negative right eigenvector w and a left eigenvector v corresponding to the zero eigenvalue.*
   Let f_k be the kth components of f such that

$$a = \sum_{k,i,j=1}^{n} v_k w_i w_j \frac{\partial^2 f_k}{\partial x_i \partial x_j}(0,0) \text{ and } b = \sum_{k,i=1}^{n} v_k w_i \frac{\partial^2 f_k}{\partial x_i \partial \beta}(0,0).$$

Th sign of a and b always determines the local dynamics of the system around the equilibrium point.

(i) $a > 0, b > 0$. *When $\beta < 0$ with $|\beta| \ll 1$; 0 is locally asymptotically stable and there exists a positive unstable equilibrium; when $0 < \beta \ll 1$; 0 is unstable and there exists a negative, locally asymptotically stable equilibrium.*

(ii) $a < 0, b < 0$. *When $\beta < 0$ with $|\beta| \ll 1$; 0 is locally asymptotically stable and there exists a positive unstable equilibrium; when $0 < \beta \ll 1$; 0 is asymptotically stable equilibrium, and there exists a positive unstable equilibrium.*

(iii) $a < 0, b < 0$. *When $\beta < 0$ with $|\beta| \ll 1$; 0 is unstable, and a positive unstable equilibrium appears.*

(iv) $a < 0, b > 0$. *When β changes from positive to negative, 0 changes its stability from stable to unstable. Correspondingly, a negative unstable equilibrium becomes positive and locally asymptotically stable.*

Particularly, if $a > 0$ and $b > 0$, then, a subcritical (or backward) bifurcation occurs at $\beta = 0$.

2.7.1. Computation of bifurcation coefficients

To determine the local dynamics of the transformed system (21), the value of a and b are computed to probe whether the model system (21) shows forward or backward bifurcation. Since $v_1 = v_2 = v_5 = v_6 = v_7 = 0$ for $k = 1; 2; 5; 6; 7$ then $k = 3; 4; 8; 9$ are considered. For the system (21), the associated non-zero second order partial derivatives at disease-free equilibrium and at $\varepsilon_2 = \varepsilon_2^*$ are given by

$$\frac{\partial^2 f_3}{\partial x_1 \partial x_9} = \frac{(1-\theta)a_1\psi\mu_h}{\pi_h(\psi+\mu_h)}, \frac{\partial^2 f_3}{\partial x_1 \partial x_4} = \frac{(1-\theta)a_2\psi\mu_h}{\pi_h(\psi+\mu_h)}, \frac{\partial^2 f_3}{\partial x_2 \partial x_9} = \frac{\theta a \mu_h}{\pi_h},$$

$$\frac{\partial^2 f_8}{\partial x_4 \partial x_6} = \frac{(1-\lambda)\varepsilon_1\omega\mu_c}{\pi_c(\omega+\mu_c)}, \frac{\partial^2 f_8}{\partial x_6 \partial x_9} = \frac{(1-\lambda)\varepsilon_2^*\psi\mu_c}{\pi_c(\omega+\mu_c)}, \frac{\partial^2 f_8}{\partial x_4 \partial x_7} = \frac{\lambda\varepsilon\mu_c}{\pi_c}.$$

Therefore,

$$a = v_3 w_1 w_9 \frac{(1-\theta)a_1\psi\mu_h}{\pi_h(\psi+\mu_h)} + v_3 w_1 w_4 \frac{(1-\theta)a_2\psi\mu_h}{\pi_h(\psi+\mu_h)} + v_3 w_2 w_9 \frac{\theta a \mu_h}{\pi_h} + v_8 w_4 w_6 \frac{(1-\lambda)\varepsilon_1\omega\mu_c}{\pi_c(\omega+\mu_c)} + v_8 w_6 w_9 \frac{(1-\lambda)\varepsilon_2^*\psi\mu_c}{\pi_c(\omega+\mu_c)} + v_8 w_4 \omega_7 \frac{\lambda\varepsilon\mu_c}{\pi_c},$$

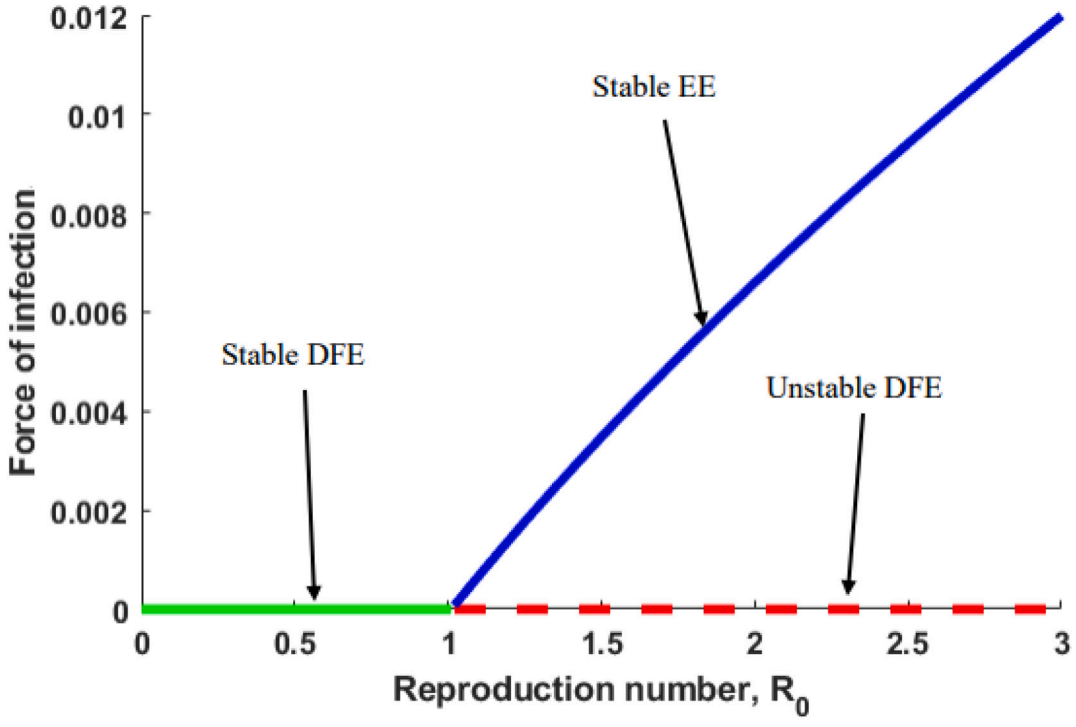


Fig. 8. Forward bifurcation diagram for the model system (1) illustrating how that the disease-free equilibrium and endemic equilibria exchange stability when $\mathcal{R}_0 = 1$. The cyan and blue curve indicate stable equilibria whereas the dashed red curve signify unstable equilibrium.

$$a = -\left[w_9\left(A_1\frac{(1-\theta)a_1\psi\mu_h}{\pi_h(\psi+\mu_h)} + A_2\frac{\theta a\mu_h}{\pi_h} + A_3\frac{(1-\lambda)\varepsilon_2^*\psi\mu_c}{\pi_c(\omega+\mu_c)}\right) + w_4\left(B_1\frac{(1-\theta)a_2\psi\mu_h}{\pi_h(\psi+\mu_h)} + B_2\frac{(1-\lambda)\varepsilon_1\omega\mu_c}{\pi_c(\omega+\mu_c)} + B_3\frac{\lambda\varepsilon\mu_c}{\pi_c}\right)\right] < 0,$$

where; $A_1 = B_1 = -\frac{\gamma_h\upsilon_4}{R_{11}R_{33}}(R_{14}w_4 + R_{19}w_9)$,

$A_2 = -\frac{\gamma_h\upsilon_4}{\mu_h R_{11}R_{33}}\left[(R_{11}R_{29} + \psi R_{19})w_9 + \psi R_{14}w_4\right]$,

$A_3 = B_2 = -\frac{\gamma_c\upsilon_9}{R_{66}R_{88}}\left[\frac{R_{64}\gamma_h}{R_{33}R_{44}}(R_{34}w_4 + R_{39}w_9) + \frac{R_{68}}{R_{88}}(R_{84}w_4 + R_{89}w_9)\right]$, $A_4 = -\frac{\gamma_h\upsilon_4}{R_{33}R_{66}}\left[\frac{R_{64}\gamma_h}{R_{33}R_{44}}(R_{34}w_4 + R_{39}w_9) + \frac{R_{68}}{R_{88}}(R_{84}w_4 + R_{89}w_9)\right]$,

$B_3 = -\frac{R_{74}\omega_4\gamma_c\upsilon_9}{\mu_c R_{88}} - \omega\left[\frac{R_{64}\gamma_h\gamma_c\upsilon_9(R_{34}\omega a_4 + R_{39}w_9)}{\mu_c R_{33}R_{44}R_{66}R_{88}} + \frac{R_{68}\gamma_c\upsilon_9(R_{84}w_4 + R_{89}w_9)}{\mu_c R_{66}R_{88}^2}\right]$.

For the value of b, it can be shown that the associated non-vanishing partial derivative is $\frac{\partial^2 f_8}{\partial x_9\partial\varepsilon_2^*} = \frac{(1-\lambda)\varepsilon_2^*\psi\mu_c}{\omega+\mu_c}$, which follows that $b = \upsilon_8 w_9\frac{(1-\lambda)\varepsilon_2^*\psi\mu_c}{\omega+\mu_c} > 0$.

Therefore, $a < 0$ and $b > 0$, hence the model system (1) exhibit a forward bifurcation.

Lemma 2.5. *The unique endemic equilibrium is guaranteed and by Theorem 4, the model is locally asymptotically stable for $\mathcal{R}_0 > 1$. In addition, by Theorem 4 item (i), the model undergoes backward bifurcation when $a > 0$. This holds only if $\upsilon_3 < 0$ and $\upsilon_8 < 0$, otherwise it undergoes forward bifurcation.*

2.8. Bifurcation diagram

Fig. 8 illustrates, the forward bifurcation (transcritical bifurcation) at $\mathcal{R}_0 = 1$ when $a < 0$ and $b > 0$ for a model system (1). Thus, the bifurcation diagram for the model system (1) illustrates that the disease-free equilibrium and endemic equilibria exchange stability when $\mathcal{R}_0 = 1$. This scenario indicates biologically that the model system (1) is globally asymptotically stable at the disease-free level.

The equilibrium point emerges when $\mathcal{R}_0 < 1$, and a unique endemic equilibrium point results whenever $\mathcal{R}_0 > 1$. The unique endemic equilibrium point is locally asymptotically stable when $\mathcal{R}_0$ is near one. Observably as $\mathcal{R}_0$ decreases below one, that is $\mathcal{R}_0 < 1$ no endemicity exists and the disease keeps waning, and as $\mathcal{R}_0$ rise to above one, that is $\mathcal{R}_0 > 1$, the disease spreads through the population.

2.9. Global stability of endemic equilibrium point

This subsection deploys the Lyapunov function method to study the global stability of endemic equilibrium point (EEP) beyond just the issue of neighbourhood equilibrium points. The Lyapunov function enables the extension of the analysis to capture an enormous region as opposed to relying only on a strip region.

Theorem 5. *If $\mathcal{R}_e > 1$, then the bovine tuberculosis disease model system (1) has a unique equilibrium point E_* which is globally asymptotically stable.*

Proof. Since a suitable Lyapunov function is developed using Korobeinikov and Wake (2002) approach, then, the Lyapunov function will be applied to analyse the global stability of the endemic equilibrium point of the model system (1).

Therefore, the Lyapunov function is developed using the general form as illustrated hereunder:

$$\begin{aligned} L = \sum_{i=1}^{9} c_i(x_i - x_i^* \ln x_i) &= c_1(x_1 - x_1^* \ln x_1) + c_2(x_2 - x_2^* \ln x_2) + c_3(x_3 - x_3^* \ln x_3) \\ &+ c_4(x_4 - x_4^* \ln x_4) + c_5(x_5 - x_5^* \ln x_5) + c_6(x_6 - x_6^* \ln x_6) + c_7(x_7 - x_7^* \ln x_7) \\ &+ c_8(x_8 - x_8^* \ln x_8) + c_9(x_9 - x_9^* \ln x_9). \end{aligned}$$

where c_i is a positive constant, x_i is the population of the ith compartment and x_i^* is the endemic equilibrium point. Thus, the following Lyapunov function is constructed:

$$\begin{aligned} L = c_1(S_{nh} - S_{nh}^* \ln S_{nh}) + c_2(S_{eh} - S_{eh}^* \ln S_{eh}) + c_3(E_h - E_h^* \ln E_h) \\ + c_4(I_h - I_h^* \ln I_h) + c_5(R_h - R_h^* \ln R_h) + c_6(S_{nv} - S_{nv}^* \ln S_{nv}) \\ + c_7(S_v - S_v^* \ln S_v) + c_8(E_c - E_c^* \ln E_c) + c_9(I_c - I_c^* \ln I_c). \end{aligned}$$

Since the chosen function L and its constant are always differentiable and continuous, the differentiating L regarding time yields the following results:

$$\begin{aligned} \frac{dL}{dt} =& c_1\left(1 - \frac{S_{nh}^*}{S_{nh}}\right)\frac{dS_{nh}}{dt} + c_2\left(1 - \frac{S_{eh}^*}{S_{eh}}\right)\frac{dS_{eh}}{dt} + c_3\left(1 - \frac{E_h^*}{E_h}\right)\frac{dE_h}{dt} \\ &+ c_4\left(1 - \frac{I_h^*}{I_h}\right)\frac{dI_h}{dt} + c_5\left(1 - \frac{R_h^*}{R_h}\right)\frac{dR_h}{dt} + c_6\left(1 - \frac{S_{nv}^*}{S_{nv}}\right)\frac{dS_{nv}}{dt} \\ &+ c_7\left(1 - \frac{S_v^*}{S_v}\right)\frac{dS_v}{dt} + c_8\left(1 - \frac{E_c^*}{E_c}\right)\frac{dE_c}{dt} + c_9\left(1 - \frac{I_c^*}{I_c}\right)\frac{dI_c}{dt}. \end{aligned} \tag{26}$$

Substituting the model equations of the model system (1) into Eq. (26) to obtain

$$\begin{aligned} \frac{dL}{dt} =& c_1\left(1 - \frac{S_{nh}^*}{S_{nh}}\right)\left(\pi_h - (\mu_h + \psi + \varphi_1)S_{nh}\right) \\ &+ c_2\left(1 - \frac{S_{eh}^*}{S_{eh}}\right)\left(\psi S_{nh} - (\mu_h + \varphi_2)S_{eh}\right) \\ &+ c_3\left(1 - \frac{E_h^*}{E_h}\right)\left(\varphi_2 S_{eh} + \varphi_1 S_{nh} - (\mu_h + \gamma_h)E_h + \tau_h R_h\right) \\ &+ c_4\left(1 - \frac{I_h^*}{I_h}\right)\left(\gamma_h E_h - (\mu_h + \alpha_h + \rho_h)I_h\right) \\ &+ c_5\left(1 - \frac{R_h^*}{R_h}\right)\left(\rho_h I_h - (\mu_h + \tau_h)R_h\right) \\ &+ c_6\left(1 - \frac{S_{nv}^*}{S_{nv}}\right)\left(\pi_c - (\mu_c + \omega + \sigma_1)S_{nv}\right) \\ &+ c_7\left(1 - \frac{S_v^*}{S_v}\right)\left(\omega S_{nv} - (\mu_c + \sigma_2)S_v\right) \\ &+ c_8\left(1 - \frac{E_c^*}{E_c}\right)\left(\sigma_1 S_{nv} + \sigma_2 S_v - (\mu_c + \gamma_c)E_c\right) \\ &+ c_9\left(1 - \frac{I_c^*}{I_c}\right)\left(\gamma_c E_c - (\mu_c + \alpha_c)I_c\right). \end{aligned} \tag{27}$$

Now, at the endemic equilibrium point of the model system (1) the following are results:

Case1 : Considering the differential equations of human population, then making $\pi_h, \psi, \tau_h, \gamma_h$ and ρ_h the subject yields the following:

$$\begin{cases} \pi_h = (\mu_h + \psi + \varphi_1^*)S_{nh}^*, \\ \psi = (\mu_h + \varphi_2^*)\dfrac{S_{eh}^*}{S_{nh}^*}, \\ \tau_h = -\dfrac{1}{R_h^*}\left(\varphi_2^* S_{eh}^* + \varphi_1^* S_{nh}^* + (\mu_h + \gamma_h)E_h^*\right), \\ \gamma_h = (\mu_h + \alpha_h + \rho_h)\dfrac{I_h^*}{E_h^*}, \\ \rho_h = (\mu_h + \tau_h)\dfrac{R_h^*}{I_h^*}. \end{cases} \tag{28}$$

Case2 : Considering the differential equations of cattle population and making π_c, ω, $(\mu_c + \gamma_c)$ and γ_c the subject yields the following:

$$\begin{cases} \pi_c = (\mu_c + \omega + \sigma_1^*)S_{nv}^*, \\ \omega = (\mu_c + \sigma_2^*)\dfrac{S_v^*}{S_{nv}^*}, \\ (\mu_c + \gamma_c) = \dfrac{1}{E_c^*}(\sigma_2^* S_v^* + \sigma_1^* S_{nv}^*), \\ \gamma_c = (\mu_h + \alpha_c)\dfrac{I_c^*}{E_c^*}. \end{cases} \tag{29}$$

Now, substituting into equation (27), then $\dfrac{dL}{dt}$ becomes:

$$\begin{aligned} \frac{dL}{dt} = {} & c_1\left(1 - \frac{S_{nh}^*}{S_{nh}}\right)\left[(\mu_h + \psi + \varphi_1^*)S_{nh}^* - (\mu_h + \psi + \varphi_1^*)S_{nh}\right] \\ & + c_2\left(1 - \frac{S_{eh}^*}{S_{eh}}\right)\left[(\mu_h + \varphi_2^*)\frac{S_{eh}^*}{S_{nh}^*}S_{nh} - (\mu_h + \varphi_2^*)S_{eh}\right] \\ & + c_3\left(1 - \frac{E_h^*}{E_h}\right)(\varphi_2 S_{eh} + \varphi_1 S_{nh} - (\mu_h + \gamma_h)E_h) \\ & - c_3\left(\frac{1}{R_h^*}\left(\varphi_2^* S_{eh}^* + \varphi_1^* S_{nh}^* + (\mu_h + \gamma_h)E_h^*\right)R_h\right) \\ & + c_4\left(1 - \frac{I_h^*}{I_h}\right)\left[(\mu_h + \alpha_h + \rho_h)\frac{I_h^*}{E_h^*}E_h - (\mu_h + \alpha_h + \rho_h)I_h\right] \\ & + c_5\left(1 - \frac{R_h^*}{R_h}\right)\left[(\mu_h + \tau_h)\frac{R_h^*}{I_h^*}I_h - (\mu_h + \tau_h)R_h\right] \\ & + c_6\left(1 - \frac{S_{nv}^*}{S_{nv}}\right)\left[(\mu_c + \omega + \sigma_1^*)S_{nv}^* - (\mu_c + \omega + \sigma_1^*)S_{nv}\right] \\ & + c_7\left(1 - \frac{S_v^*}{S_v}\right)\left[(\mu_c + \sigma_2^*)\frac{S_v^*}{S_{nv}^*}S_{nv} - (\mu_c + \sigma_2)S_v\right] \\ & + c_8\left(1 - \frac{E_c^*}{E_c}\right)\left[\sigma_1 S_{nv} + \sigma_2 S_v - \frac{1}{E_c^*}(\sigma_2^* S_v^* + \sigma_1^* S_{nv}^*)E_c\right] \\ & + c_9\left(1 - \frac{I_c^*}{I_c}\right)\left[(\mu_h + \alpha_c)\frac{I_c^*}{E_c^*}E_c - (\mu_c + \alpha_c)I_c\right]. \end{aligned} \tag{30}$$

To investigate the global stability of the endemic equilibrium using the Lyapunov function approach at the steady state, set $\dfrac{dN_h^*}{dt} = 0$ and $\dfrac{dN_c^*}{dt} = 0$, such that:

$$\frac{dN_h^*}{dt} = \pi_h - \mu_h N_h^* - \alpha_h\ I_h^* = 0 \implies N_h^* = \frac{\pi_h - \alpha_h\ I_h^*}{\mu_h}, \tag{31}$$

and

$$\frac{dN_c^*}{dt} = \pi_c - \mu_c N_c^* - \alpha_c\ I_c^* = 0 \implies N_c^* = \frac{\pi_c - \alpha_c\ I_c^*}{\mu_c}. \tag{32}$$

Substituting equation (31) and (32) into forces of infections to obtain:

$$\begin{aligned} \varphi_1^* &= \frac{\mu_h(1-\theta)}{\pi_h - \alpha_h\ I_h^*}(a_1 I_c^* + a_2 I_h^*),\ \varphi_2^* = a\theta\frac{\mu_h I_c^*}{\pi_h - \alpha_h\ I_h^*}. \\ \sigma_1^* &= \frac{\mu_c(1-\lambda)}{\pi_c - \alpha_c\ I_c^*}(\varepsilon_1 I_h^* + \varepsilon_2 I_c^*),\ \sigma_2^* = \varepsilon\lambda\frac{\mu_c I_h^*}{\pi_c - \alpha_c\ I_c^*}. \end{aligned} \tag{33}$$

After substituting equation (33) into (30), the equation becomes:

$$\begin{aligned}
\frac{dL}{dt} &= c_1\left(1-\frac{S^*_{nh}}{S_{nh}}\right)\left[\left(\mu_h+\psi+\frac{\mu_h(1-\theta)(a_1I^*_c+a_2I^*_h)}{\pi_h-\alpha_h\ I^*_h}\right)S^*_{nh}\right.\\
&\quad\left.-\left(\mu_h+\psi+\frac{\mu_h(1-\theta)(a_1I^*_c+a_2I^*_h)}{\pi_h-\alpha_h\ I^*_h}\right)S_{nh}\right]\\
&\quad+c_2\left(1-\frac{S^*_{eh}}{S_{eh}}\right)\left[\left(\mu_h+\frac{a\theta\mu_hI^*_c}{\pi_h-\alpha_h\ I^*_h}\right)\frac{S^*_{eh}}{S^*_{nh}}S_{nh}-\left(\mu_h+\frac{a\theta\mu_hI^*_c}{\pi_h-\alpha_h\ I^*_h}\right)S_{eh}\right]\\
&\quad+c_3\left(1-\frac{E^*_h}{E_h}\right)\left(\frac{a\theta\mu_hI^*_c}{\pi_h-\alpha_h\ I^*_h}\right)S_{eh}+\frac{\mu_h(1-\theta)(a_1I^*_c+a_2I^*_h)}{\pi_h-\alpha_h\ I^*_h}S_{nh}-(\mu_h+\gamma_h)E_h\\
&\quad-\left(\frac{1}{R^*_h}\left(\frac{a\theta\mu_hI^*_c}{\pi_h-\alpha_h\ I^*_h}S^*_{eh}+\frac{\mu_h(1-\theta)(a_1I^*_c+a_2I^*_h)}{\pi_h-\alpha_h\ I^*_h}S^*_{nh}+(\mu_h+\gamma_h)E^*_h\right)R_h\right)\\
&\quad+c_4\left(1-\frac{I^*_h}{I_h}\right)\left[(\mu_h+\alpha_h+\rho_h)\frac{I^*_h}{E^*_h}E_h-(\mu_h+\alpha_h+\rho_h)I_h\right]\\
&\quad+c_5\left(1-\frac{R^*_h}{R_h}\right)\left[(\mu_h+\tau_h)\frac{R^*_h}{I^*_h}I_h-(\mu_h+\tau_h)R_h\right]\\
&\quad+c_6\left(1-\frac{S^*_{nv}}{S_{nv}}\right)\left[\left(\mu_c+\omega+\frac{\mu_c(1-\lambda)(\varepsilon_1I^*_h+\varepsilon_2I^*_c)}{\pi_c-\alpha_c\ I^*_c}\right)S^*_{nv}\right.\\
&\quad\left.-\left(\mu_c+\omega+\frac{\mu_c(1-\lambda)(\varepsilon_1I^*_h+\varepsilon_2I^*_c)}{\pi_c-\alpha_c\ I^*_c}\right)S_{nv}\right]\\
&\quad+c_7\left(1-\frac{S^*_v}{S_v}\right)\left[\left(\mu_c+\frac{\varepsilon\lambda\mu_cI^*_h}{\pi_c-\alpha_c\ I^*_c}\right)\frac{S^*_v}{S^*_{nv}}S_{nv}-\left(\mu_c+\frac{\varepsilon\lambda\mu_cI^*_h}{\pi_c-\alpha_c\ I^*_c}\right)S_v\right]\\
&\quad+c_8\left(1-\frac{E^*_c}{E_c}\right)\left[\frac{\mu_c(1-\lambda)(\varepsilon_1I^*_h+\varepsilon_2I^*_c)}{\pi_c-\alpha_c\ I^*_c}S_{nv}+\frac{\varepsilon\lambda\mu_cI^*_h}{\pi_c-\alpha_c\ I^*_c}S_v\right.\\
&\quad\left.-\frac{1}{E^*_c}\left(\frac{\varepsilon\lambda\mu_cI^*_h}{\pi_c-\alpha_c\ I^*_h}S^*_v+\frac{\mu_c(1-\lambda)(\varepsilon_1I^*_h+\varepsilon_2I^*_c)}{\pi_c-\alpha_c\ I^*_c}S^*_{nv}\right)E_c\right]\\
&\quad+c_9\left(1-\frac{I^*_c}{I_c}\right)\left[(\mu_h+\alpha_c)\frac{I^*_c}{E^*_c}E_c-(\mu_c+\alpha_c)I_c\right].
\end{aligned}\tag{34}$$

Simplifying equation (34) it gives;

$$\begin{aligned}
\frac{dL}{dt} &= c_1\left(1-\frac{S^*_{nh}}{S_{nh}}\right)\left[(\mu_h+\psi)S^*_{nh}-(\mu_h+\psi)S_{nh}\right]\\
&\quad+c_6\left(1-\frac{S^*_{nv}}{S_{nv}}\right)\left[(\mu_c+\omega+)S^*_{nv}-(\mu_c+\omega)S_{nv}\right]\\
&\quad+\Phi(P)
\end{aligned}\tag{35}$$

$$\frac{dL}{dt} = -c_1(\mu_h+\psi)\left(1-\frac{S^*_{nh}}{S_{nh}}\right)^2S_{nh}-c_6(\mu_c+\omega)\left(1-\frac{S^*_{nv}}{S_{nv}}\right)^2S_{nv}+\Phi(P)\tag{36}$$

where $P=(S_{nh},S_{eh},E_h,I_h,R_h,S_{nv},S_v,E_c,I_c)$ such that P is non negative. Therefore

$$\begin{aligned}
\Phi(P) &= -c_1\left(\frac{\mu_h(1-\theta)(a_1I^*_c+a_2I^*_h)}{\pi_h-\alpha_h\ I^*_h}\right)\left(1-\frac{S^*_{nh}}{S_{nh}}\right)^2S_{nh}\\
&\quad-c_2\left(\mu_h+\frac{a\theta\mu_hI^*_c}{\pi_h-\alpha_hI^*_h}\right)\left(1-\frac{S^*_{eh}}{S_{eh}}\right)\left(1-\frac{S^*_{eh}S_{nh}}{S_{eh}S^*_{nh}}\right)S_{eh}\\
&\quad-c_3\left(\frac{\mu_h(1-\theta)(a_1I^*_c+a_2I^*_h)}{\pi_h-\alpha_h\ I^*_h}\right)\left(1-\frac{E^*_h}{E_h}\right)\left(1-\frac{S_{nh}}{S^*_{nh}}\right)\frac{R_h}{R^*_h}S^*_{nh}\\
&\quad-c_3\left(\frac{a\theta\mu_hI^*_c}{\pi_h-\alpha_h\ I^*_h}\right)\left(1-\frac{E^*_h}{E_h}\right)\left(1-\frac{S_{nh}}{S^*_{nh}}\right)\frac{R_h}{R^*_h}S^*_{nh}
\end{aligned}$$

$$
\begin{aligned}
&- c_3(\mu_h+\gamma_h)\left(1-\frac{E_h^*}{E_h}\right)^2\frac{R_h}{R_h^*}E_h\\
&- c_4(\mu_h+\alpha_h+\rho_h)\left(1-\frac{I_h^*}{I_h}\right)\left(1-\frac{I_h^*E_h}{I_hE_h^*}\right)I_h\\
&- c_5(\mu_h+\tau_h)\left(1-\frac{R_h^*}{R_h}\right)\left(1-\frac{R_h^*I_h}{R_hI_h^*}\right)R_h\\
&- c_6\left(\frac{\mu_c(1-\lambda)(\varepsilon_1 I_h^*+\varepsilon_2 I_c^*)}{\pi_c-\alpha_c\ I_c^*}\right)\left(1-\frac{S_{nv}^*}{S_{nv}}\right)^2 S_{nv}\\
&- c_7\left(\mu_c+\frac{\varepsilon\lambda\mu_c I_h^*}{\pi_c-\alpha_c\ I_c^*}\right)\left(1-\frac{S_v^*}{S_v}\right)\left(1-\frac{S_v^*S_{nv}}{S_vS_{nv}^*}\right)S_v\\
&- c_8\left(\frac{\mu_c(1-\lambda)(\varepsilon_1 I_h^*+\varepsilon_2 I_c^*)}{\pi_c-\alpha_c\ I_c^*}\right)\left(1-\frac{E_c^*}{E_c}\right)\left(1-\frac{E_cS_{nv}^*}{E_c^*S_{nv}}\right)E_cS_{nv}\\
&- c_8\left(\frac{\varepsilon\lambda\mu_c I_h^*}{\pi_c-\alpha_c\ I_c^*}\right)\left(1-\frac{E_c^*}{E_c}\right)\left(1-\frac{S_v^*}{E_c^*S_v}\right)S_vE_c\\
&- c_9(\mu_h+\alpha_c)\left(1-\frac{I_c^*}{I_c}\right)\left(1-\frac{I_c^*E_c}{I_cE_c^*}\right)I_c.
\end{aligned}
\tag{37}
$$

Following the approach of Korobeinikov and Wake (2002), the function $\Phi(S_{nh}, S_{eh}, E_h, I_h, R_h, S_{nv}, S_v, E_c, I_c)$ is shown to be non-negative, i.e., $\Phi \leq 0$ for all states in the domain. This implies that the time derivative of the Lyapunov function satisfies $\frac{dL}{dt} \leq 0$, and equality holds only at the endemic equilibrium point E_*. By LaSalle's Invariance Principle (La Salle, 1976), the largest invariant set where $\frac{dL}{dt} = 0$ is the singleton $\{E_*\}$, implying that E_* is globally asymptotically stable in the interior of the feasible region Ω. This result indicates that if the system is perturbed from the endemic state, it will eventually return to it over time. Therefore, the disease persists at a constant endemic level in both humans and cattle populations, and eradication is not possible under these conditions. The population will stabilize around this endemic equilibrium regardless of the initial level of infection, provided the parameters remain unchanged. □

2.10. Parameter estimation via MCMC

To validate the bTB model (1), human and cattle data collected between 2016 and 2019 were used for parameter estimation. The dataset comprised time series for recovered humans, exposed cattle, and infectious cattle. A Bayesian Markov Chain Monte Carlo (MCMC) approach was employed to estimate model parameters and assess their identifiability. This method incorporates parameter uncertainty by sampling from posterior distributions that align model predictions with observed data. An adaptive Metropolis algorithm was utilized to dynamically refine the proposal distribution during sampling, thereby improving convergence and computational efficiency.

A subset of parameters was fitted to the observed data to ensure accurate characterization of the bTB transmission dynamics. These fitted parameters include the recovery rate of infectious humans (ρ_h), the rate of waning immunity among recovered humans (τ_h), progression rates of exposed humans and cattle to the infectious stage (γ_h, γ_c), disease-induced mortality rates in humans and cattle (α_h, α_c), effectiveness of public health education campaigns (θ), vaccination effectiveness among cattle (λ), and the per capita education rate among non-educated humans (ψ). The estimation process was conducted within the Bayesian MCMC framework using the 2016–2019 dataset, ensuring that model outputs closely matched the empirical observations while accounting for parameter uncertainty through posterior distributions and the estimates are presented in Table 1.

Figs. 9–11 compare the model-generated trajectories with observed data for the recovered human, exposed cattle, and infectious cattle compartments during the 2016–2019 study period. The strong agreement between simulated and empirical trends demonstrates that the model effectively captures the epidemiological patterns of bTB transmission. Specifically, Fig. 9 illustrates the increasing trend in recovered human cases, consistent with the influence of treatment and recovery efforts. Fig. 10 depicts the temporal dynamics of exposed cattle, reflecting variations in infection pressure and control measures. Fig. 11 shows the decline in infectious cattle cases towards the end of the study period, likely attributable to improved vaccination coverage and awareness interventions. Overall, these findings confirm that the model (1) provides a reliable representation of the transmission dynamics between humans and cattle.

2.11. Optimal control formulation

Optimal control theory is employed to determine the best way to minimize the spread bTB in both human and cattle population. At end, the study specifically seeks to determine how the combination of the public health education campaign, treatment, and vaccination efforts could be harnessed to best reduce the spread of the bTB strain. The extension of model (1) treated the intervention strategies as functions of time rather than parameters. These control measures are presented as $u_1(t) \in [0, 1]$ that is the public health education campaign for protecting susceptible human from consuming contaminated cattle products, coming into contact with infectious cattle and inter-human infections, $u_2(t) \in [0, 1]$ refers to human treatment that minimizes contracting the disease from infected humans and $u_3(t) \in [0, 1]$ denotes the effort devoted to the vaccination of cattle in a bid to reduce the number of infections from infectious human, contaminated environment and inter-cattle infections to finally eradicate the

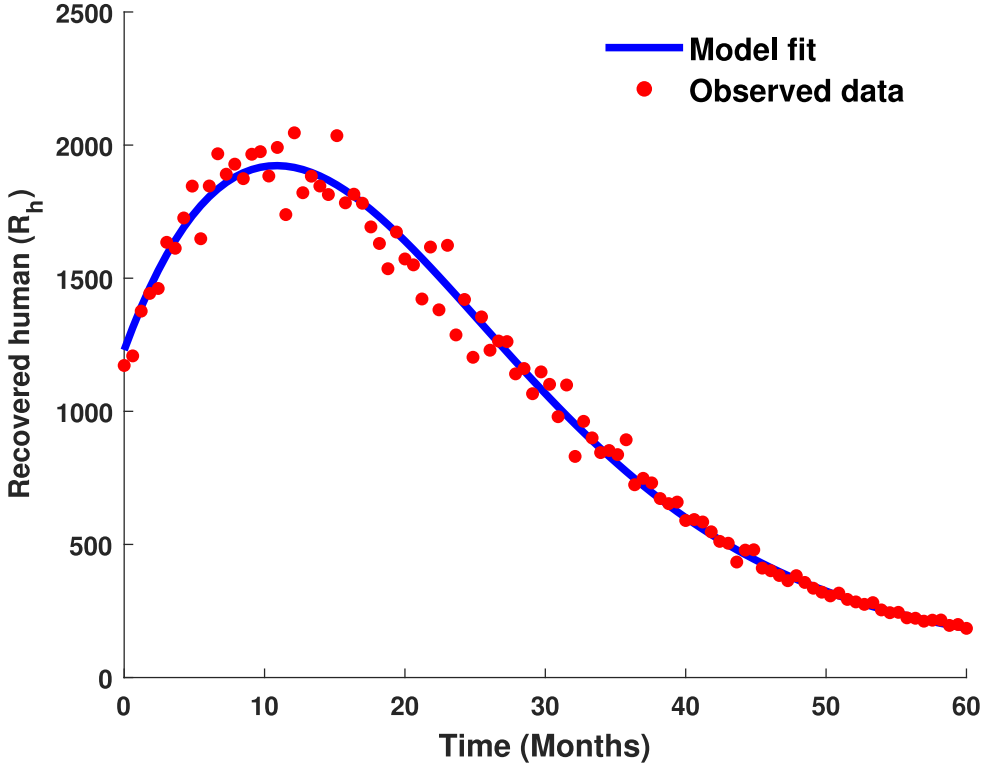


Fig. 9. Recovered human.

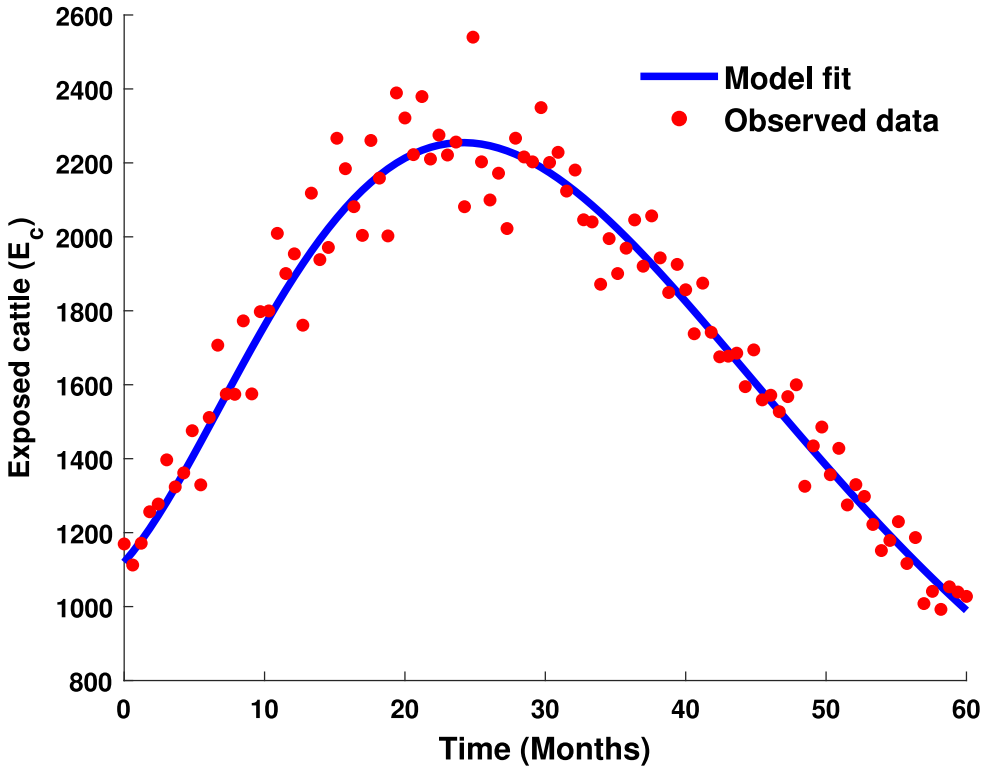


Fig. 10. Exposed cattle.

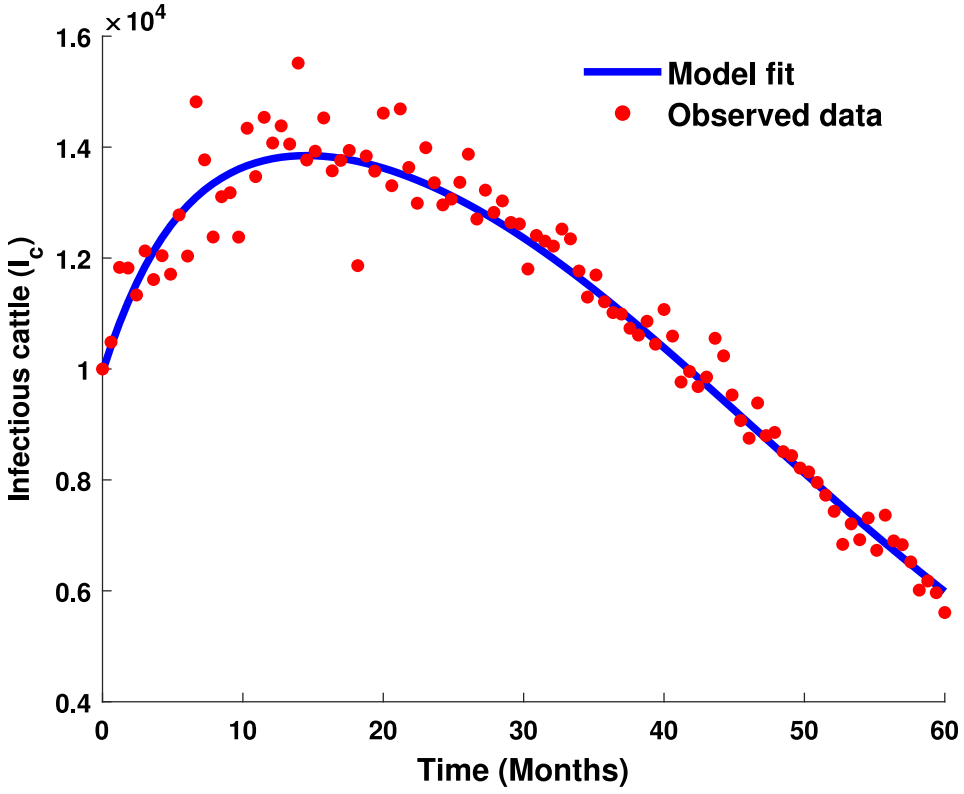


Fig. 11. Infectious cattle.

disease in the localized area. As such, the modified model becomes

$$\begin{cases} \frac{dS_{nh}}{dt} &= \pi_h - (\mu_h + \varphi_1)S_{nh} - \psi u_1 S_{nh}, \\ \frac{dS_{eh}}{dt} &= \psi u_1 S_{nh} - \mu_h S_{eh} - \varphi_2(1-u_1)S_{eh}, \\ \frac{dE_h}{dt} &= \varphi_1 S_{nh} + \varphi_2(1-u_1)S_{eh} - (\mu_h + \gamma_h)E_h + \tau_h R_h, \\ \frac{dI_h}{dt} &= \gamma_h E_h - (\mu_h + \alpha_h) - \rho_h u_2 I_h, \\ \frac{dR_h}{dt} &= \rho_h u_2 I_h - (\mu_h + \tau_h)R_h, \\ \frac{dS_{nv}}{dt} &= \pi_c - (\mu_c + \sigma_1)S_{nv} - \omega u_3 S_{nv}, \\ \frac{dS_v}{dt} &= \omega u_3 S_{nv} - \mu_c S_v - \sigma_2(1-u_3)S_v, \\ \frac{dE_c}{dt} &= \sigma_1 S_{nv} + \sigma_2(1-u_3)S_v - (\mu_c + \gamma_c)E_c, \\ \frac{dI_c}{dt} &= \gamma_c E_c - (\mu_c + \alpha_c)I_c, \end{cases} \tag{38}$$

whereby, $\varphi_1 = (1-\theta)\left(a_1\frac{I_c}{N_h} + a_2\frac{I_h}{N_h}\right)$, $\varphi_2 = a\theta\frac{I_c}{N_h}$, $\sigma_1 = (1-\lambda)\left(\varepsilon_1\frac{I_h}{N_c} + \varepsilon_2\frac{I_c}{N_c}\right)$, $\sigma_2 = \varepsilon\lambda\frac{I_h}{N_c}$ and the corresponding initial conditions are $S_{nh}(0) > 0$, $S_{eh}(0) > 0$, $E_h(0) \geq 0$, $I_h(0) \geq 0$, $R_h(0) \geq 0$, $S_{nv}(0) > 0$, $S_v(0) > 0$, $E_c(0) \geq 0$, $I_c(0) \geq 0$.

Since the aim is to limit the number of infectious human and cattle to the lowest number, then for the model (38) the objective functional for minimization will have the form $J = \min_{(u_1,u_2,u_3)} \int_0^{t_f} \left(A_1 I_h + A_2 I_c + \frac{1}{2}\sum_{i=1}^{3} c_i u_i^2\right) dt$. where t_f is the final time and A_1, A_2 represent the associated costs that minimize the number of infected human and infected cattle respectively. The term $\sum_{i=1}^{3}\frac{c_i}{2}u_i^2$ represents the cost related to a particular control strategy, that is, $\frac{1}{2}c_1u_1^2$, $\frac{1}{2}c_2u_2^2$ and $\frac{1}{2}c_3u_3^2$ which represent the costs of executing public health education campaigns, treatment and vaccination, respectively as discussed in the works of Mlay et al. (2015), Mlay and Hugo (2021), Ndendya et al. (2025d, 2024), Butcher (1996).

In this regard, Lenhart and Workman (2007) proved that the control variable u_i for $i = 1, 2, 3$ are given in quadratic form because, if they take a linear form it becomes difficult to solve for u_i^* using the optimality condition. Moreover, in case of the higher non-linear control variable such as polynomials of degree three and above, the Hamiltonian function may fail to produce the optimal value. Therefore, to minimize the objective functional J, the optimal control pair (u_1^*, u_2^*, u_3^*) is required constructed as $J(u_1^*, u_2^*, u_3^*) = \min\{J(u_1, u_2, u_3) : (u_1, u_2, u_3) \in \xi\}$, whereby the control set ξ refers to the Lebesgue measurable aspect, which is defined as $\xi = \{(u_1, u_2, u_3) : 0 \leq u_i \leq 1,\ t \in [0, t_f],\ \text{for } i = 1, 2, 3\ \}$.

Theorem 6. *With regard to an optimal control problem (38), there is an optimal control pair* $(u_1^*, u_2^*, u_2^*) \in \xi$ *such that* $J(u_1^*, u_2^*, u_2^*) = min\{J(u_1, u_2, u_3) : (u_1, u_2, u_3) \in \xi\}$ *provided that the following five conditions are met.*

1. *Set of controls and state variables are non-empty.*
2. *Control set* ξ *is convex and closed.*
3. *The right-hand side of the model system (38) is bounded by a linear system in the state and control variables.*
4. *The integrand* $I(x_i, u_i)$ *of the objective functional is convex.*
5. *There are positive numbers* $n_1, n_2 > 0$ *and* $r > 1$ *such that the integrand* $I(x_i, u_i)$ *of the objective functional satisfies* $I(x_i, u_i) \geq n_1 \left(\sum_{i=1}^{3} |u_i|^2 \right)^{\frac{r}{2}} - n_2$.

Proof. According to Lukes (1982), Michael et al. (2025), Ndendya Jufren Zakayo and Liana Yustina Amon (2025a), Ndendya Jufren Zakayo et al. (2025i), Ndendya Jufren Zakayo and Liana Yustina Amon (2025b), Ndendya Jufren et al. (2025c), the results from the solutions of the model system (38) are non-empty, since the set of control strategies and state variables are positive which verify the first condition. Moreover, using the definition of convex set, the control set $(u_1, u_2, u_3) \in \xi$ is closed and convex hence validating the second condition. Furthermore, the model system (38) is bounded which invites a conclusion to the effect the third condition has been met. The integrand in the objective functional which is $I(x_i, u_i) = A_1 I_h + A_2 I_c + \frac{1}{2} \sum_{i=1}^{3} c_i u_i^2$ is convex on control set ξ which supports the validity of the fourth condition. Finally, the positive numbers exist such that $n_1, n_2 > 0$ and $r > 1$ in which $A_1 I_h + A_2 I_c + \frac{1}{2} \sum_{i=1}^{3} c_i u_i^2 \geq n_1 \left(\sum_{i=1}^{3} |u_i|^2 \right)^{\frac{r}{2}} - n_2$. The bounded state variables complete and confirm the existence of an optimal control of the model system (38). Therefore, there exist an optimal control pair $(u_1^*, u_2^*, u_2^*) \in \xi$ which minimizes the objective functional J. □

2.12. Characterization of the optimal controls

This section describes the application of the Pontryagin's Maximum Principle to convert the model system (38) and the objective functional into the problem for minimizing a piecewise Hamiltonian (H) function with respect to control variables u_1, u_2 and u_3. The Pontryagin's Maximum Principle also facilitates the derivation of the conditions that are essential for the optimal control pair (Lenhart and Workman, 2007; Mlay et al., 2015; Ndendya et al., 2023; Ndendya Jufren Zakayo et al., 2025a).

Now, we define the Hamiltonian function (H) as follows,

$$\begin{aligned} H = {} & A_1 I_h + A_2 I_c + \frac{1}{2} c_1 u_1^2 + \frac{1}{2} c_2 u_2^2 + \frac{1}{2} c_3 u_3^2 + \varpi_1 \frac{dS_{nh}}{dt} + \varpi_2 \frac{dS_{eh}}{dt} + \varpi_3 \frac{dE_h}{dt} + \varpi_4 \frac{dI_h}{dt} \\ & + \varpi_5 \frac{dR_h}{dt} + \varpi_6 \frac{dS_{nv}}{dt} + \varpi_7 \frac{dS_v}{dt} + \varpi_8 \frac{dE_c}{dt} + \varpi_9 \frac{dI_c}{dt}, \end{aligned} \tag{39}$$

whereby, $A_1 I_h + A_2 I_c + \frac{1}{2} c_1 u_1^2 + \frac{1}{2} c_2 u_2^2 + \frac{1}{2} c_3 u_3^2$ is the integrand of the objective functional (J) and ϖ_i, $i = 1, 2, \ldots, 9$ are adjoint (co-state) variable introduced for appending the differential equations with the objective functional (J).

Hence, expanding the Hamiltonian function (H) in Eq. (39) yields,

$$\begin{aligned} H = {} & A_1 I_h + A_2 I_c + \frac{1}{2} c_1 u_1^2 + \frac{1}{2} c_2 u_2^2 + \frac{1}{2} c_3 u_3^2 \\ & + \varpi_1 \left(\pi_h - \mu_h S_{nh} - (1-\theta) \left(a_1 \frac{I_c}{N_h} + a_2 \frac{I_h}{N_h} \right) S_{nh} - \psi u_1 S_{nh} \right) \\ & + \varpi_2 \left(\psi u_1 S_{nh} - \mu_h S_{eh} - (1-u_1) a\theta \frac{I_c}{N_h} S_{eh} \right) \\ & + \varpi_3 \left((1-\theta) \left(a_1 \frac{I_c}{N_h} + a_2 \frac{I_h}{N_h} \right) S_{nh} + (1-u_1) a\theta \frac{I_c}{N_h} S_{eh} - (\mu_h + \gamma_h) E_h + \tau_h R_h \right) \\ & + \varpi_4 \left(\gamma_h E_h - (\mu_h + \alpha_h) I_h - \rho_h u_2 I_h \right) \\ & + \varpi_5 \left(\rho_h u_2 I_h - (\mu_h + \tau_h) R_h \right) \\ & + \varpi_6 \left(\pi_c - \mu_c S_{nv} - (1-\lambda) \left(\varepsilon_1 \frac{I_h}{N_c} + \varepsilon_2 \frac{I_c}{N_c} \right) S_{nv} - \omega u_3 S_{nv} \right) \\ & + \varpi_7 \left(\omega u_3 S_{nv} - \mu_c S_v - (1-u_3) \varepsilon \lambda \frac{I_h}{N_c} S_v \right) \\ & + \varpi_8 \left((1-\lambda) \left(\varepsilon_1 \frac{I_h}{N_c} + \varepsilon_2 \frac{I_c}{N_c} \right) S_{nv} + (1-u_3) \varepsilon \lambda \frac{I_h}{N_c} S_v - (\mu_c + \gamma_c) E_c \right) \\ & + \varpi_9 \left(\gamma_c E_c - (\mu_c + \alpha_c) I_c \right). \end{aligned} \tag{40}$$

Now, recalling the idea behind the Pontryagin's Maximum Principle and the existence of an optimal control point translates into Theorem 7.

Theorem 7. *For an optimal control* $(u_1^*, u_2^*, u_2^*) \in \xi$ *and the corresponding set of optimal solution* $(S_{nh}^*, S_{eh}^*, E_h^*, I_h^*, R_h^*, S_{nv}^*, S_v^*, E_c^*, I_c^*)$ *that minimizes the objective functional* (J)*, there exist co-state variables (adjoint variable)* $\varpi_1, \varpi_2, \varpi_3, \varpi_4, \varpi_5, \varpi_6, \varpi_7, \varpi_8, \varpi_9$ *such that,*

$$\begin{cases} \dfrac{d\varpi_i}{dt} = -\dfrac{\partial H}{\partial x_i} & (\textit{adjoint conditions}), \\ \varpi_i(t_f) = 0 & (\textit{transversality conditions}), \\ \dfrac{\partial H}{\partial u_i} = 0 & \textit{at } u_i^* \quad (\textit{optimality conditions}). \end{cases} \tag{41}$$

Proof. To get the adjoint system, the Hamiltonian function as in Eq. (40) is differentiated partially with respect to state variables ($S_{nh}, S_{eh}, E_h, I_h, R_h, S_{nv}, S_v, E_c, I_c$) respectively which gives,

$$\begin{aligned}
\frac{d\varpi_1}{dt} &= -\left(\frac{\partial H}{\partial S_{nh}}\right) = \mu_h\varpi_1 + (\varpi_1-\varpi_2)\psi u_1 + (\varpi_1-\varpi_3)(1-\theta)\left(a_1\frac{I_c}{N_h} + a_2\frac{I_h}{N_h}\right),\\
\frac{d\varpi_2}{dt} &= -\left(\frac{\partial H}{\partial S_{eh}}\right) = \mu_h\varpi_2 + (\varpi_2-\varpi_3)(1-u_1)a\theta\frac{I_c}{N_h},\\
\frac{d\varpi_3}{dt} &= -\left(\frac{\partial H}{\partial E_h}\right) = \mu_h\varpi_3 + (\varpi_3-\varpi_4)\gamma_h,\\
\frac{d\varpi_4}{dt} &= -\left(\frac{\partial H}{\partial I_h}\right) = -A_1 + (\varpi_1-\varpi_3)(1-\theta)a_2\frac{S_{nh}}{N_h} + (\varpi_4-\varpi_5)\rho_h u_2 + (\mu_h+\alpha_h)\varpi_4 +\\
&\qquad (\varpi_6-\varpi_8)(1-\lambda)\varepsilon_1\frac{S_{nv}}{N_c} + (\varpi_7-\varpi_8)(1-u_3)\lambda\varepsilon\frac{S_v}{N_c},\\
\frac{d\varpi_5}{dt} &= -\left(\frac{\partial H}{\partial R_h}\right) = \mu_h\varpi_5 + (\varpi_5-\varpi_3)\tau_h,\\
\frac{d\varpi_6}{dt} &= -\left(\frac{\partial H}{\partial S_{nv}}\right) = \mu_c\varpi_6 + (\varpi_6-\varpi_7)\omega u_3 + (\varpi_6-\varpi_8)(1-\lambda)\left(\varepsilon_1\frac{I_h}{N_c} + \varepsilon_2\frac{I_c}{N_c}\right),\\
\frac{d\varpi_7}{dt} &= -\left(\frac{\partial H}{\partial S_v}\right) = \mu_c\varpi_7 + (\varpi_7-\varpi_8)(1-u_3)\varepsilon\lambda\frac{I_h}{N_c},\\
\frac{d\varpi_8}{dt} &= -\left(\frac{\partial H}{\partial E_c}\right) = \mu_c\varpi_8 + (\varpi_8-\varpi_9)\gamma_c,\\
\frac{d\varpi_9}{dt} &= -\left(\frac{\partial H}{\partial I_c}\right) = -A_2 + (\varpi_1-\varpi_3)(1-\theta)a_1\frac{S_{nh}}{N_h} + (\varpi_6-\varpi_8)(1-\lambda)\varepsilon_2\frac{S_{nv}}{N_c} +\\
&\qquad (\mu_c+\alpha_c)\varpi_9.
\end{aligned} \tag{42}$$

With the final time conditions/transversality conditions ($t_f = T$), $\varpi_1(T) = \varpi_2(T) = \varpi_3(T) = \varpi_4(T) = \varpi_5(T) = \varpi_6(T) = \varpi_7(T) = \varpi_8(T) = \varpi_9(T) = 0$.

Furthermore, applying the optimality conditions, to obtain the optimal control (u_1^*, u_2^*, u_3^*) such that,

$$\frac{\partial H}{\partial u_i} = 0 \implies \frac{\partial H}{\partial u_1} = \frac{\partial H}{\partial u_2} = \frac{\partial H}{\partial u_3} = 0, \tag{43}$$

so that,

$$\begin{aligned}
&\frac{\partial H}{\partial u_1} = c_1u_1 + (\varpi_2-\varpi_1)\psi S_{nh} + (\varpi_2-\varpi_3)\theta a\frac{I_c}{N_h}S_{eh},\\
&\frac{\partial H}{\partial u_1} = c_1u_1^* + (\varpi_2-\varpi_1)\psi S_{nh} + (\varpi_2-\varpi_3)\theta a\frac{I_c}{N_h}S_{eh} = 0, \quad \text{at} \quad u_1 = u_1^*\\
&\implies u_1^*(t) = \frac{1}{c_1}(\varpi_1-\varpi_2)\psi S_{nh} + \frac{1}{c_1}(\varpi_3-\varpi_2)\theta a\frac{I_c}{N_h}S_{eh}.
\end{aligned} \tag{44}$$

$$\begin{aligned}
&\frac{\partial H}{\partial u_2} = c_2u_2 + (\varpi_5-\varpi_4)\rho_h I_h\\
&\frac{\partial H}{\partial u_2} = c_2u_2^* + (\varpi_5-\varpi_4)\rho_h I_h = 0 \quad \text{at} \quad u_2 = u_2^*\\
&\implies u_2^*(t) = \frac{1}{c_2}(\varpi_4-\varpi_5)\rho_h I_h.
\end{aligned} \tag{45}$$

$$\begin{aligned}
&\frac{\partial H}{\partial u_3} = c_3u_3 + (\varpi_7-\varpi_6)\omega S_{nv} + (\varpi_7-\varpi_8)\lambda\varepsilon\frac{I_h}{N_c}S_v,\\
&\frac{\partial H}{\partial u_3} = c_3u_3^* + (\varpi_7-\varpi_6)\omega S_{nv} + (\varpi_7-\varpi_8)\lambda\varepsilon\frac{I_h}{N_c}S_v = 0, \quad \text{at} \quad u_3 = u_3^*\\
&\implies u_3^*(t) = \frac{1}{c_3}(\varpi_6-\varpi_7)\omega S_{nv} + \frac{1}{c_3}(\varpi_8-\varpi_7)\lambda\varepsilon\frac{I_h}{N_c}S_v.
\end{aligned} \tag{46}$$

Since the controls are bounded, including the bounds of the control from (44)–(46) gives,

$$u_1^* = \begin{cases}
0 \quad \text{if} \quad \frac{1}{c_1}(\varpi_1-\varpi_2)\psi S_{nh} + \frac{1}{c_1}(\varpi_3-\varpi_2)\theta a\frac{I_c}{N_h}S_{eh} \le 0,\\
\frac{1}{c_1}(\varpi_1-\varpi_2)\psi S_{nh} + \frac{1}{c_1}(\varpi_3-\varpi_2)\theta a\frac{I_c}{N_h}S_{eh} \quad \text{if}\\
\quad 0 < \frac{1}{c_1}(\varpi_1-\varpi_2)\psi S_{nh} + \frac{1}{c_1}(\varpi_3-\varpi_2)\theta a\frac{I_c}{N_h}S_{eh} < 1,\\
1 \quad \text{if} \quad \frac{1}{c_1}(\varpi_1-\varpi_2)\psi S_{nh} + \frac{1}{c_1}(\varpi_3-\varpi_2)\theta a\frac{I_c}{N_h}S_{eh} \ge 1.
\end{cases} \tag{47}$$

$$u_2^* = \begin{cases} 0 & \text{if} \quad \frac{1}{c_2}(\varpi_4 - \varpi_5)\rho_h I_h \le 0, \\ \frac{1}{c_2}(\varpi_4 - \varpi_5)\rho_h I_h & \text{if} \quad 0 < \frac{1}{c_2}(\varpi_4 - \varpi_5)\rho_h I_h < 1, \\ 1 & \text{if} \quad \frac{1}{c_2}(\varpi_4 - \varpi_5)\rho_h I_h \ge 1. \end{cases} \tag{48}$$

$$u_3^* = \begin{cases} 0 & \text{if} \quad \frac{1}{c_3}(\varpi_6 - \varpi_7)\omega S_{nv} + \frac{1}{c_3}(\varpi_8 - \varpi_7)\lambda\varepsilon\frac{I_h}{N_c}S_v \le 0, \\ \frac{1}{c_3}(\varpi_6 - \varpi_7)\omega S_{nv} + \frac{1}{c_3}(\varpi_8 - \varpi_7)\lambda\varepsilon\frac{I_h}{N_c}S_v & \text{if} \\ \quad 0 < \frac{1}{c_3}(\varpi_6 - \varpi_7)\omega S_{nv} + \frac{1}{c_3}(\varpi_8 - \varpi_7)\lambda\varepsilon\frac{I_h}{N_c}S_v < 1, \\ 1 & \text{if} \quad \frac{1}{c_3}(\varpi_6 - \varpi_7)\omega S_{nv} + \frac{1}{c_3}(\varpi_8 - \varpi_7)\lambda\varepsilon\frac{I_h}{N_c}S_v \ge 1. \end{cases} \tag{49}$$

The control set can equivalently be written as,

$$\begin{aligned} u_1^* &= \min\left\{1, \max\left\{0, \frac{1}{c_1}(\varpi_1 - \varpi_2)\psi S_{nh} + \frac{1}{c_1}(\varpi_3 - \varpi_2)\theta a\frac{I_c}{N_h}S_{eh}\right\}\right\}, \\ u_2^* &= \min\left\{1, \max\left\{0, \frac{1}{c_2}(\varpi_4 - \varpi_5)\rho_h I_h\right\}\right\}, \\ u_3^* &= \min\left\{1, \max\left\{0, \frac{1}{c_3}(\varpi_6 - \varpi_7)\omega S_{nv} + \frac{1}{c_3}(\varpi_8 - \varpi_7)\lambda\varepsilon\frac{I_h}{N_c}S_v\right\}\right\}. \end{aligned} \tag{50}$$

□

3. Numerical simulations and discussion

Numerical simulations of the optimal control system is conducted by coupling the state and adjoint equations using the classical fourth-order Runge–Kutta method as discussed in Ndendya et al. (2025e), Liana Yustina Amon et al. (2025), Ndendya Jufren Zakayo et al. (2025b), Ashgi et al. (2021), Iskandar and Tiong (2022), Lede and Mungkasi (2019), Musa et al. (2010). The analysis focuses on evaluating the effectiveness of various intervention strategies in minimizing bovine tuberculosis (bTB) transmission among human and cattle populations. The forward simulation of the state equations employs initial conditions: $S_{nh} = 15000$, $S_{eh} = 1000$, $E_h = 1000$, $I_h = 500$, $R_h = 300$, $S_{nv} = 15000$, $S_v = 10$, $E_c = 6000$, and $I_c = 10$. In contrast, the backward integration of the adjoint equations which is performed using terminal conditions $\varpi_i(T) = 0$ over a time horizon of $T = 60$ months. The objective functional J is evaluated using arbitrarily chosen weight constants: $A_1 = 20$, $A_2 = 50$, $c_1 = 200$, $c_2 = 100$, and $c_3 = 50$, to demonstrate the qualitative behaviour of the optimal control solutions. Parameter values used in the simulations are listed in Table 1. A total of seven distinct control scenarios are explored to assess and compare the impact of individual and combined control strategies on disease mitigation.

3.1. Strategy A: Single method control with only education campaign

Fig. 12 illustrates the simulation results for Scenario A, where public health education is implemented as the sole control intervention. As shown in Fig. 12(a), the number of infected humans declines significantly due to the widespread dissemination of educational campaigns. A similar downward trend is observed in the infected cattle population, as depicted in Fig. 12(b), attributed indirectly to the reduced human-to-cattle transmission. The control profile associated with this intervention, $u_1(t)$, is presented in Fig. 12(c), indicating that maximum effort (100%) should be sustained for the first 59 months, followed by a sharp decline to zero at the final time point, $T = 60$ months.

3.2. Strategy B: Single method control with only treatment

Fig. 12(a) and Fig. 12(b) demonstrate that implementing treatment as a standalone intervention significantly accelerates the reduction of infections in both human and cattle populations, compared to the absence of control measures. The corresponding control profile for u_2, shown in Fig. 12(c), indicates that a sustained 100% treatment effort is required for the first 58 months, followed by a sharp decline approaching zero at the final time point $T = 60$ months.

3.3. Strategy C: Single method control with only vaccination

As illustrated in Figs. 13(a) and 13(b), vaccination is a key intervention in reducing the number of bTB infections in both human and cattle populations. The control profile for u_3, depicted in Fig. 13(c), indicates that the vaccination effort should be sustained at 100% for approximately 59 months before tapering off to zero.

3.4. Strategy D: Double methods control using both education campaign and treatment

The simulation results for the combined implementation of public health education (u_1) and treatment (u_2) strategies, shown in Figs. 14(a) and 14(b), demonstrate a significant reduction in the number of infected humans and cattle. The corresponding control profiles in Fig. 14(c) indicate that u_1 should be maintained at its maximum level for 52 months, while u_2 remains at full effort for 19 months before both decline to zero.

3.5. Strategy E: Double methods control using both education campaign and vaccination

Control scenario E demonstrates a substantial reduction in the number of infected humans and cattle, as confirmed by Figs. 15(a) and 15(b). The control profile in Fig. 15(c) indicates that the education campaign (u_1) should be sustained at full intensity for 53 months before sharply declining to zero, while vaccination (u_3) should be maintained at 100% effort for approximately 57 months prior to its decline.

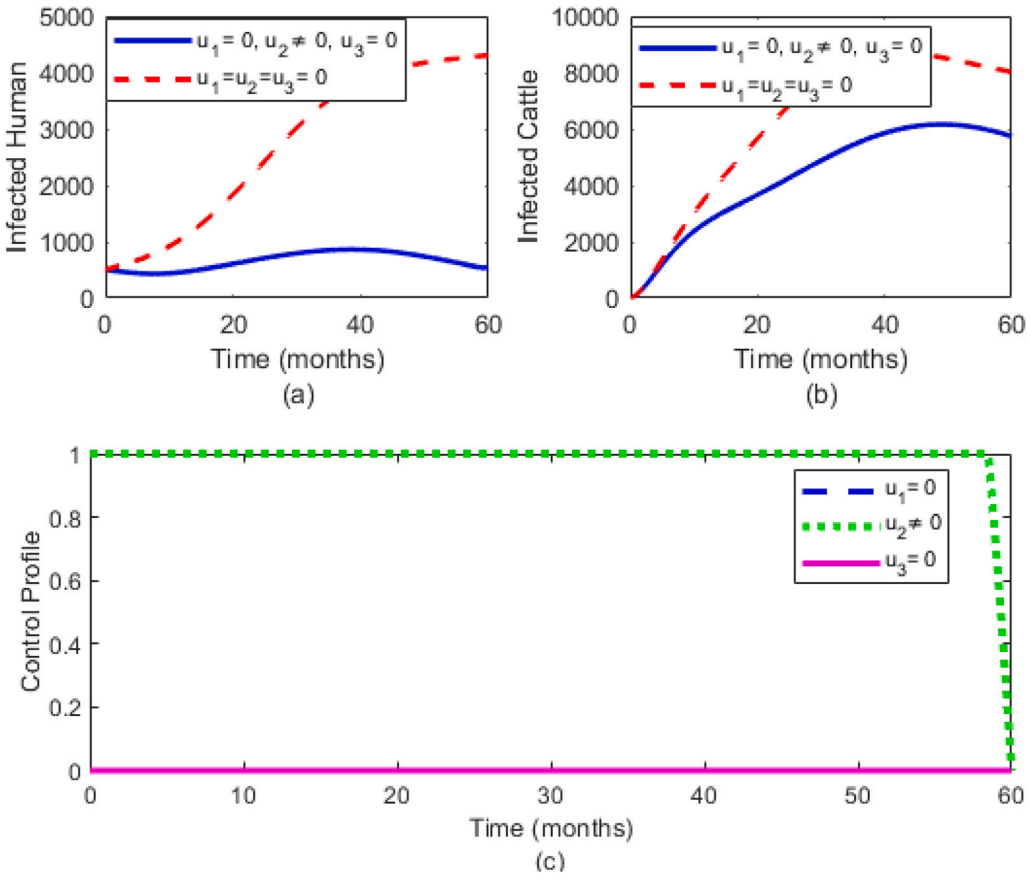


Fig. 12. Effects of treatment on bTB transmission dynamics.

3.6. Strategy F: Double method control using both treatment and vaccination

In this scenario, the combined implementation of treatment (u_2) and vaccination (u_3), with education (u_1) set to zero, aims to further reduce infection incidence. Simulation results shown in Figs. 16(a) and 16(b) demonstrate a significant decrease in infected humans and cattle due to this dual intervention strategy. Moreover, the control profile in Fig. 16(c) indicates that maintaining u_2 and u_3 at full intensity for 59 and 55 months, respectively, is necessary before both controls can be gradually reduced.

3.7. Strategy G: Triple method control using the education campaign, treatment and vaccination

Finally, the simulation results for the combined application of all three intervention strategies are presented in Figs. 17(a) and 17(b). These results demonstrate a notably greater reduction in the number of infected humans and cattle compared to any previous dual-strategy scenarios. Furthermore, the control profiles in Fig. 17(c) show that controls u_1, u_2, and u_3 should be maintained at full intensity (100%) for 5, 19, and 26 months, respectively, before gradually decreasing towards zero by the end of the intervention period.

3.8. Cost-effectiveness analysis

Cost-effectiveness analysis is a critical methodology for identifying the most economical intervention to mitigate disease transmission within resource-constrained populations. This approach evaluates competing control strategies by comparing their respective costs and health outcomes, emphasizing the additional cost per incremental health benefit (Ndendya et al., 2023; Ndendya Jufren Zakayo et al., 2025a; Okosun et al., 2013; Okosun, 2020). The Incremental Cost-Effectiveness Ratio (ICER) is employed to determine the optimal strategy to minimize the spread of bovine tuberculosis (bTB). The ICER is defined as the ratio of the difference in intervention costs to the difference in the total number of infections averted between two strategies, expressed as $ICER = \frac{\text{Cost}_L - \text{Cost}_N}{\text{Effect}_L - \text{Effect}_N} = \frac{\Delta C}{\Delta E}$, where L and N denote two control strategies under comparison, C represents the total intervention cost, and E is the health outcome quantified by the number of infections averted. Here, ΔC and ΔE correspond to the incremental cost and incremental effect, respectively.

The total number of infections averted, E, is calculated by subtracting the cumulative infections under control from the total infections expected without intervention, given by:

$$E = T_f \left(I_h(0) + I_c(0)\right) - \int_0^{T_f} \left(I_h^*(t) + I_c^*(t)\right) dt, \tag{51}$$

where $(I_h^*(t), I_c^*(t))$ denote the optimal infectious human and cattle populations under the optimal controls (u_1^*, u_2^*, u_3^*), and $I_h(0) + I_c(0)$ represents the initial infection load in the absence of control interventions (i.e., $u_1 = u_2 = u_3 = 0$).

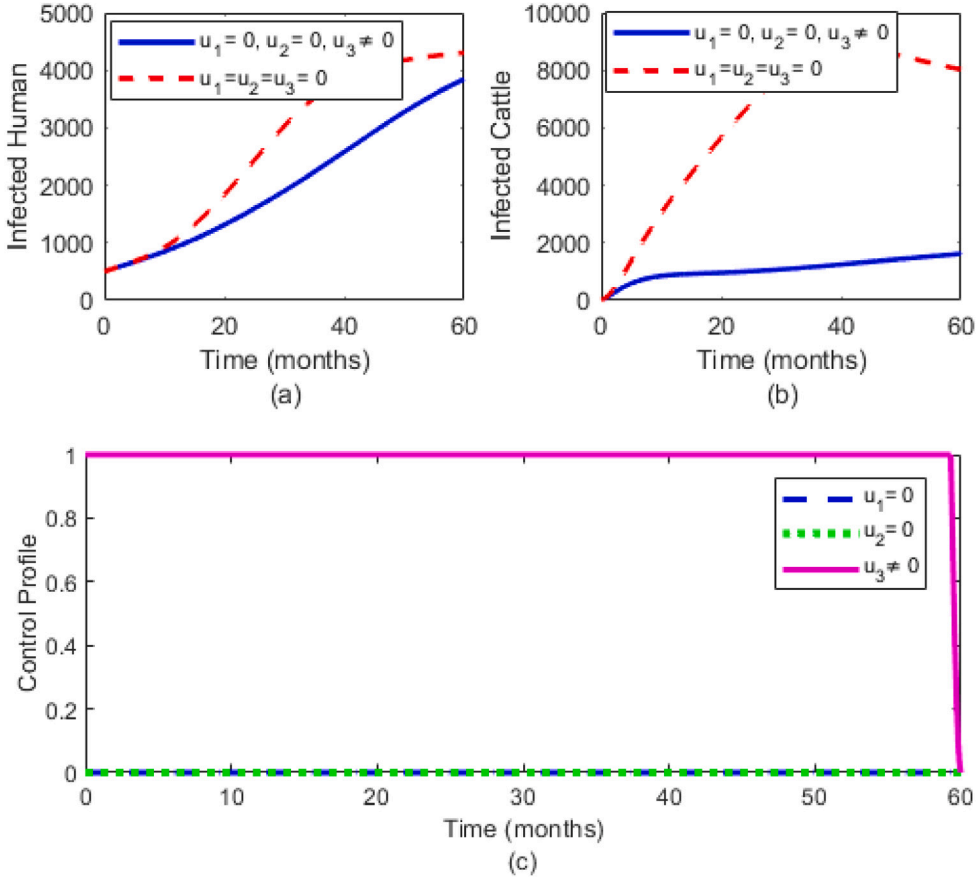


Fig. 13. Effects of vaccination on bTB transmission dynamics.

Table 2
Number of infections averted and total cost of each strategy.

Strategies	Infections (E)	Infection averted(E)	Total cost ($) ($C$)
No control	4992 800	–	0
Strategy A	3136 400	1856 400	440.77
Strategy B	2665 800	2327 000	15 643
Strategy C	1679 200	3313 600	331.47
Strategy D	1082 100	3910 700	46 395
Strategy E	855 790	4137 010	776.72
Strategy F	404 780	4588 020	87 581
Strategy G	344 020	4648 780	48 154

The total cost, C, associated with each control strategy over the intervention horizon $[0, T_f]$ is defined as:

$$C = \int_0^{T_f} \left(B_1 u_1 S_{nh}(t) + B_2 u_2 I_h(t) + B_3 u_3 S_{nv}(t) \right) dt, \tag{52}$$

where B_1, B_2, and B_3 represent the unit costs of implementing public health education (u_1), treatment of infectious humans (u_2), and vaccination of susceptible cattle (u_3), respectively. For the purposes of this study, these costs were assigned the values $B_1 = 0.3$, $B_2 = 0.5$, and $B_3 = 0.02$.

The system of equations (38), (42), and (50) was simulated for each control strategy scenario. The results, arranged in ascending order of infections averted, are summarized in Table 2, facilitating the identification of the most cost-effective interventions.

In the absence of control strategy, the total number of infectious individuals is 4992800 whereas the total cost is 0 (see Table 2). For example, the use of control strategy C implies the sum of infectious individual is 1679200, and the total cost is USD 331.47. Then, finding the number of infections averted (E) entails determining difference between the total number of infectious cases without control and the total number of infectious with control (Total number of infection − Total number of susceptibility cases prevented) (i.e, 4992800 − 1679200 = 3313600). Implies that the infection averted for strategy C is 3313600, and the same procedures are essential in determining the number of infections averted for the remaining strategies, as Table 3 illustrates:

Calculating and comparing the ICER for control strategies A and B is done as follows: ICER(A) = $\frac{\Delta C}{\Delta E} = \frac{440.77}{1856400} = 0.00024\$$ and ICER(B) = $\frac{\Delta C}{\Delta E} = \frac{15643}{2327000} = 0.03230\$$. The comparison between control measures A and B indicate that ICER(B) > ICER(A), suggesting that strategy B is costlier but less effective than strategy A. As a result, omitting strategy B from the set of alternatives, and instead comparing strategies A and C (see Table 4).

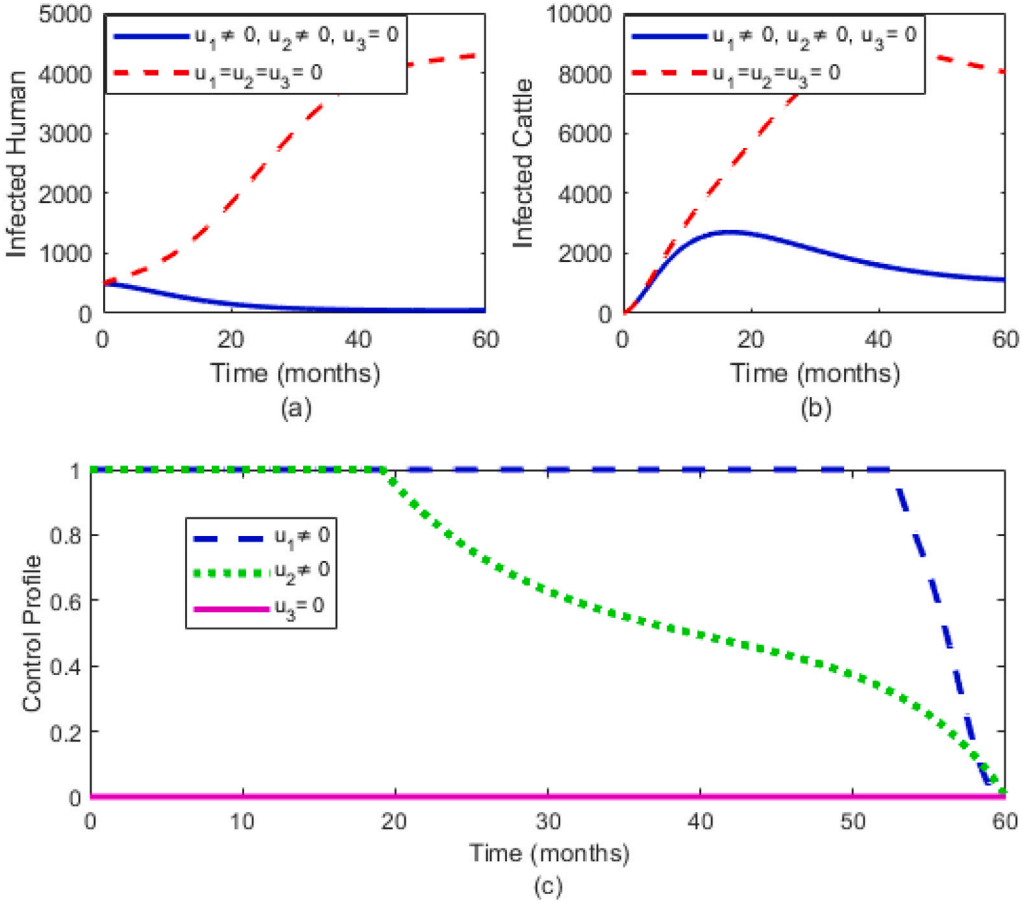


Fig. 14. Effects of dual public health education campaign and treatment control on bTB transmission dynamics.

Table 3
Incremental Cost-Effectiveness Ratio of different optimal control strategies.

Strategies	Inf. averted (E)	ΔE	C($)	ΔC($)	ICER(ΔC/ΔE)
Strategy A	1856 400	1856 400	440.77	440.77	0.00024
Strategy B	2327 000	470 600	15 643	15 202.23	0.03230
Strategy C	3313 600	986 600	331.47	−15311.53	−0.01551
Strategy D	3910 700	597 100	46 395	46 063.53	0.07714
Strategy E	4137 010	226 310	776.72	−45618.28	−0.20157
Strategy F	4588 020	451 010	87 581	86 804.28	0.19247
Strategy G	4648 780	60 760	48 154	−39427	−0.64890

Table 4
Comparing strategies A and C exclusive of strategy B.

Strategies	Inf. averted (E)	ΔE	C($)	ΔC($)	ICER(ΔC/ΔE)
Strategy A	1856 400	1856 400	440.77	440.77	0.00024
Strategy C	3313 600	1457 200	331.47	−109.3	−0.00007
Strategy D	3910 700	597 100	46 395	46063.53	0.07714
Strategy E	4137 010	226 310	776.72	−45618.28	−0.20157
Strategy F	4588 020	451 010	87 581	86804.28	0.19247
Strategy G	4648 780	60 760	48 154	−39427	−0.64890

Calculations and comparison of ICER for control strategy A and C is realized as:

$\text{ICER(A)} = \frac{\Delta C}{\Delta E} = \frac{440.77}{1856400} = 0.00024\$$ and $\text{ICER(C)} = \frac{\Delta C}{\Delta E} = \frac{-109.3}{1457200} = -0.00007\$$. Comparing control strategy A and C reveals ICER(A) > ICER(C). The negative sign for ICER C implies that strategy A is strongly dominated and less effective than strategy C. In other words, strategy A is much more costly but less effective than strategy C. As such, we omit strategy A from the set of alternatives, and instead comparing strategies C and D.

Based on Table 5, to calculate and compare the ICER for control strategies C and D results to:

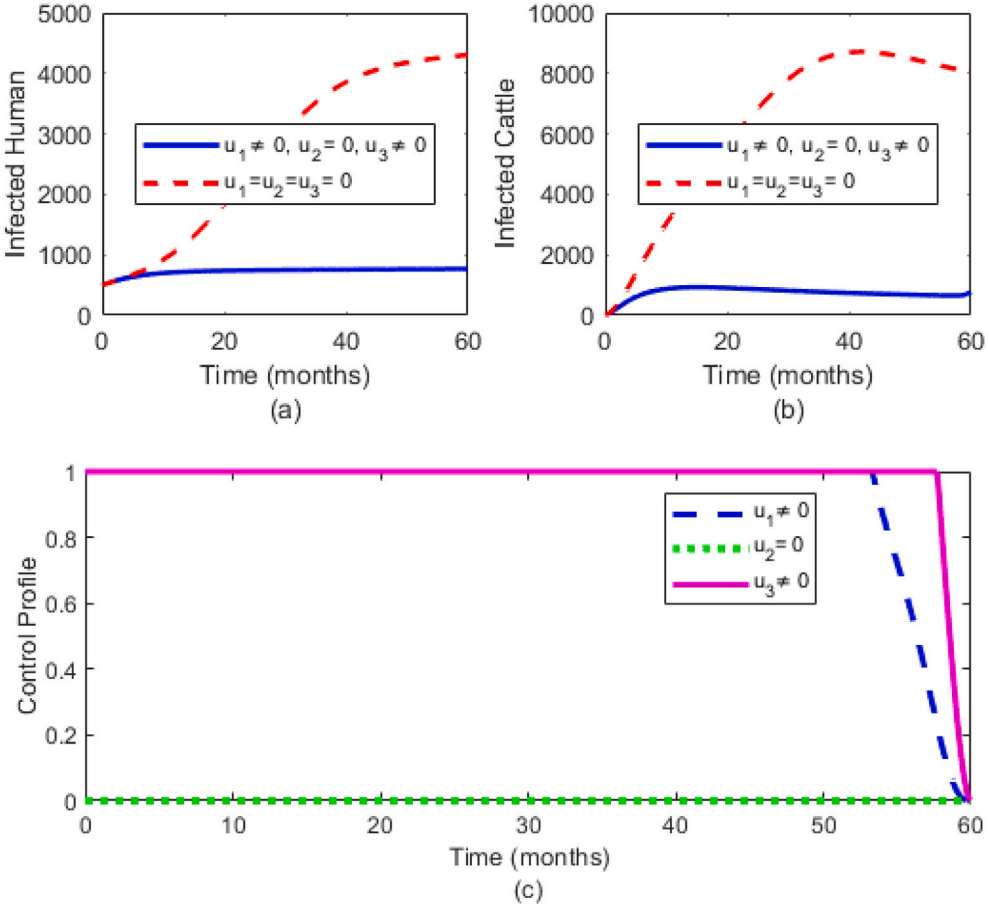


Fig. 15. Effects of dual public education campaign and vaccination control on bTB transmission dynamics.

Table 5
Comparing strategies C and D exclusive strategy A.

Strategies	Inf. averted (E)	ΔE	C($)	ΔC($)	ICER(ΔC/ΔE)
Strategy C	3313 600	3313 600	331.47	331.47	0.00010
Strategy D	3910 700	597 100	46 395	46063.53	0.07714
Strategy E	4137 010	226 310	776.72	−45618.28	−0.20157
Strategy F	4588 020	451 010	87 581	86804.28	0.19247
Strategy G	4648 780	60 760	48 154	−39427	−0.64890

Table 6
Comparing strategies C and E exclusive strategy D.

Strategies	Inf. averted (E)	ΔE	C($)	ΔC($)	ICER(ΔC/ΔE)
Strategy C	3313 600	3313 600	331.47	331.47	0.00010
Strategy E	4137 010	823 410	776.72	445.25	0.00054
Strategy F	4588 020	451 010	87 581	86804.28	0.19247
Strategy G	4648 780	60 760	48 154	−39427	−0.64890

$\text{ICER(C)} = \frac{\Delta C}{\Delta E} = \frac{331.47}{3313600} = 0.00010\$$ and $\text{ICER(D)} = \frac{\Delta C}{\Delta E} = \frac{46063.53}{597100} = 0.07714\$$. Comparing control strategies C and D reveals ICER(D) > ICER(C). The lower ICER C signals that strategy D is strongly dominated and less effective than strategy C. Thus, to strategy D from the set of alternatives and compare measures C and E instead.

Based on Table 6, to calculate and compare the ICER for control strategies C and E as follows:

$\text{ICER(C)} = \frac{\Delta C}{\Delta E} = \frac{331.47}{3313600} = 0.00010\$$ and $\text{ICER(E)} = \frac{\Delta C}{\Delta E} = \frac{445.25}{823410} = 0.00054\$$. Comparing control strategies C and E indicate ICER(E) > ICER(C). In other words, strategy E is more costly and less effective than strategy C. As such, drop strategy E from the set of alternatives, and instead, compare the costs of strategies C and F.

Using data from Table 7, we Calculate and compare the respective ICER for control strategies C and E as follows:

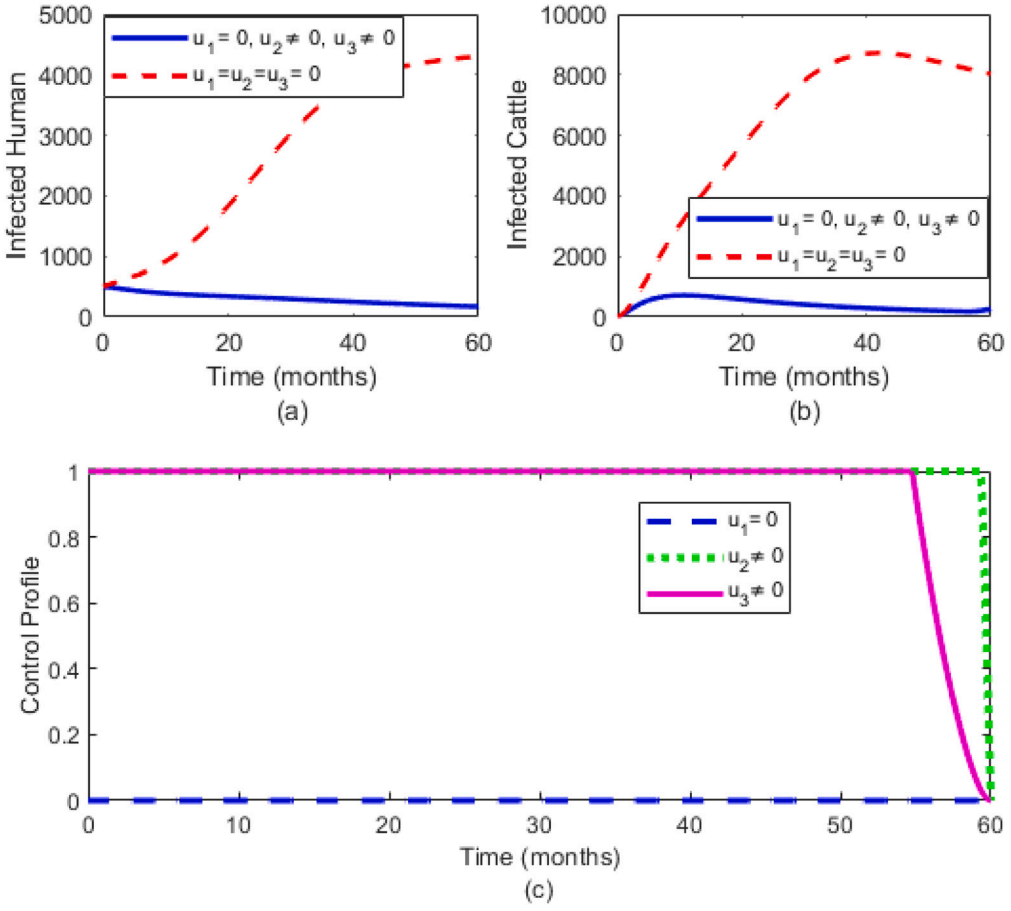


Fig. 16. Effects of treatment and vaccination on bTB transmission dynamics.

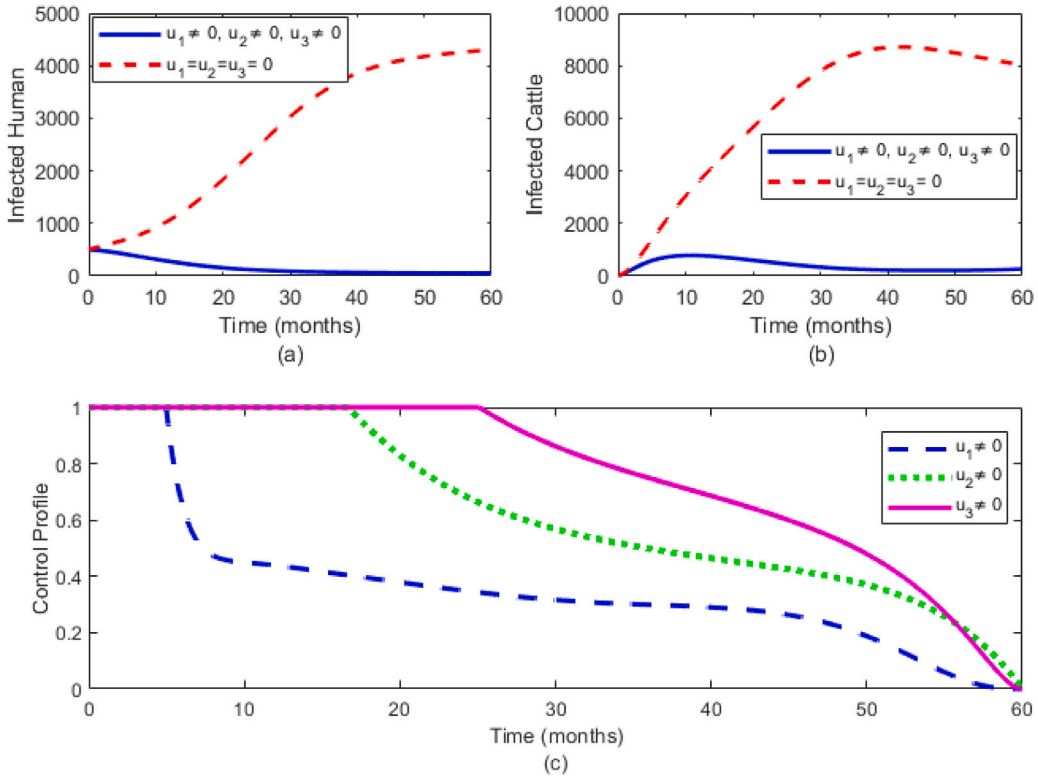


Fig. 17. Effects of triple public health education campaign, treatment and vaccination control on bTB transmission dynamics.

Table 7
Comparing strategies C and F exclusive strategy E.

Strategies	Inf. averted (E)	ΔE	C($)	ΔC($)	ICER(ΔC/ΔE)
Strategy C	3313 600	3313 600	331.47	331.47	0.00010
Strategy F	4588 020	1274 420	87 581	87249.53	0.06846
Strategy G	4648 780	60 760	48 154	−39427	−0.64890

Table 8
Comparing strategies C and G exclusive strategy F.

Strategies	Inf. averted (E)	ΔE	C($)	ΔC($)	ICER(ΔC/ΔE)
Strategy C	3313 600	3313 600	331.47	331.47	0.00010
Strategy G	4648 780	1335 180	48 154	47822.53	0.03582

$\text{ICER(C)} = \frac{\Delta C}{\Delta E} = \frac{331.47}{3313600} = 0.00010\$$ and $\text{ICER(F)} = \frac{\Delta C}{\Delta E} = \frac{87249.53}{1274420} = 0.06846\$$. This comparing control strategy C and F indicates that ICER(F) > ICER(C). In other words, strategy F is more costly and less effective than strategy C, a fact that results in the exclusion of strategy F from the set of alternatives so as to compare strategies C and G.

Based on Table 8, calculation and comparison of the ICER for control strategies C and G is done as follows:

$\text{ICER(C)} = \frac{\Delta C}{\Delta E} = \frac{331.47}{3313600} = 0.00010\$$ and $\text{ICER(F)} = \frac{\Delta C}{\Delta E} = \frac{47822.53}{1335180} = 0.03582\$$.

The comparative analysis of control strategies, based on the Incremental Cost-Effectiveness Ratio (ICER), demonstrates that Strategy C, cattle vaccination is the most cost-effective approach for mitigating bovine tuberculosis (bTB). Strategy G, which combines education, treatment, and vaccination, while achieving slightly higher infection reduction, incurs significantly greater costs and a higher ICER, making it less economically efficient than vaccination alone. Consequently, Strategy C is identified as the preferred intervention, especially in resource-constrained settings. Furthermore, while public health education contributes to reducing transmission, and treatment offers short-term benefits, neither matches the long-term impact and cost-effectiveness of vaccination. These findings reinforce the prioritization of vaccination programs as the cornerstone of bTB control policies.

4. Conclusion and recommendation

Using human and cattle data from Tanzania (2016–2019), this study developed a time-dependent optimal control model to investigate bovine tuberculosis (bTB) transmission and assess intervention strategies including public health education, treatment, and vaccination. The effective reproduction number, $\mathcal{R}_e$, was derived, confirming that bTB can be eradicated when $\mathcal{R}_e < 1$, with stability analysis verifying the disease-free equilibrium. Optimal control conditions were established via Pontryagin's Maximum Principle, and numerical simulations were conducted using forward–backward fourth-order Runge–Kutta schemes. Results indicate that integrated strategies, particularly combining vaccination, treatment, and education, achieved the greatest reductions in human and cattle infections.

Cost-effectiveness analysis identified cattle vaccination as the most economically efficient single intervention, averting over 3.3 million infections at the lowest cost and minimal ICER. While vaccination alone is highly cost-effective, integrated interventions provide additional epidemiological benefits in high-risk or high-transmission areas, albeit at increased implementation costs. These findings highlight the importance of prioritizing vaccination programs while supplementing them with treatment and public health education, especially in endemic regions of Tanzania.

Successful implementation requires strengthened veterinary infrastructure, cold-chain systems, trained personnel, sustainable funding, and multisectoral coordination among health and livestock authorities to enable surveillance, data sharing, and community engagement as suggested by Centers for Disease Control and Prevention (2012), Ayele et al. (2004), Aldila et al. (2019). Continuous epidemiological monitoring and model-informed evaluations are recommended to refine intervention strategies and maintain cost-effectiveness over time.

Study limitations include reliance on available surveillance data, potential underreporting, simplifications in transmission pathways, and the exclusion of stochastic, spatial, and behavioural heterogeneities. Future research should incorporate stochastic modelling, fractional calculus to capture memory effects, spatial heterogeneity, and individual-based simulations to enhance model realism and support context-specific, sustainable strategies for the control and eventual eradication of bTB (Oswald Sylas et al., 2025a; Ndendya Jufren Zakayo et al., 2025g; Mumbu et al., 2025; Ndendya Jufren Zakayo et al., 2025h; Elisamehe Bright et al., 2025; Nyerere Neema and Edward Stephen, 2025).

CRediT authorship contribution statement

Sylas Oswald: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Theresia Crispin Marijani:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Goodluck Mika Mlay:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Winifrida Kidima:** Writing – review & editing, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Funding statement

This work was supported indirectly by existing research funding, which facilitated the preparation and completion of the study, although the funds were not allocated specifically for this project.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

I, Sylas Oswald, gratefully acknowledge Prof. Eunice Mureithi, Prof. Berge Tsanou, and Prof. Michael Chapwanya for securing the fund that enabled this work, which was conducted at the Department of Mathematics, University of Dar es Salaam, as part of the Central and Eastern Africa Female Health-Oriented Modeling Epidemics Consortium (CAF-HOMEC) Project, generously supported by the Bill & Melinda Gates Foundation (Grant No. INV-022584). The authors also wish to sincerely thank the anonymous reviewers for their insightful comments and suggestions, which significantly improved the quality of this work.

Data availability

The data used to estimate the model parameters are detailed in Table 1. Whoever interested in accessing the dataset can request it by contacting the corresponding author via email: sylasossy2023@gmail.com.

References

Al-Obeidi, A.S., Al-Azzawi, S.F., 2022. A novel six-dimensional hyperchaotic system with self-excited attractors and its chaos synchronisation. Int. J. Comput. Sci. Math. 15 (1), 72–84.

Aldila, D., Latifah, S.L., Dumbela, P.A., 2019. Dynamical analysis of mathematical model for Bovine Tuberculosis among human and cattle population. Commun. Biomath Sci. 2 (1), 55–64.

Ashgi, R., Pratama, M.A.A., Purwani, S., 2021. Comparison of numerical simulation of epidemiological model between Euler method with 4th order Runge Kutta method. Int. J. Glob. Oper. Res. 2 (1), 37–44.

Ayele, W.Y., Neill, S.D., Zinsstag, J., Weiss, M.G., Pavlik, I., 2004. Bovine tuberculosis: an old disease but a new threat to Africa. Int. J. Tuberc. Lung Dis. 8 (8), 924–937.

Bronsvoort, B.M., et al., 2010. Cost-effectiveness of cattle vaccination against bTB in Cameroon. Prev. Vet. Med..

Butcher, J.C., 1996. A history of Runge-Kutta methods. Appl. Numer. Math. 20 (3), 247–260.

de Camino-Beck, T., Lewis, M.A., 2007. A new method for calculating net reproductive rate from graph reduction with applications to the control of invasive species. Bull. Math. Biol. 69, 1341–1354.

de Camino-Beck, T., Lewis, M.A., 2008. On net reproductive rate and the timing of reproductive output. Amer. Nat. 172 (1), 128–139.

de Camino-Beck, T., Lewis, M.A., Van den Driessche, P., 2009. A graph-theoretic method for the reproduction number in continuous time epidemiological models. J. Math. Biol. 503–516.

Castillo-Chavez, C., Feng, Z., Huang, W., 2002. In: Castillo-Chavez, C., Blower, S., van den Driessche, P., Kirschner, D., Yakubu, A.A. (Eds.), On the Computational of Ro and Its Role of Global Stability. Springer.

Castillo-Chavez, C., Song, B., 2004. Dynamical models of tuberculosis and their applications. Math. Biosci. Eng. 1 (2), 361.

Centers for Disease Control and Prevention, 2012. Mycobacterium bovis (bovine tuberculosis) in humans. Center for Disease Control and Prevention factsheets.

Clifford, D.L., Kazwala, R.R., Sadiki, H., Roug, A., Muse, E.A., Coppolillo, P.C., Mazet, J.A.K., 2022. Tuberculosis infection in wildlife from the ruaha ecosystem, Tanzania: implications for wildlife, domestic animals, and human health. EcoHealth 19, 155–164.

Diallo, B., Okelo, J.A., Osman, S., Karanja, S., Aguegboh, N.S., 2023. A study of fractional bovine tuberculosis model with vaccination on human population. Commun. Math. Biol. Neurosci..

Diallo, M., et al., 2024. Fractional-order model for the evolution of Bovine Tuberculosis with vaccination and contaminated environment. Adv. Math. Biosci. 1–18.

Diekmann, O., Heesterbeek, J.A.P., Metz, J.A., 1990. On the definition and the computation of the basic reproduction ratio R0 in models for infectious diseases in heterogeneous populations. J. Math. Biol. 28 (4), 365–382.

Dumont, Y., Chiroleu, F., Domerg, C., 2008. On a temporal model for the chikungunya disease: modeling, theory and numerics. Math. Biosci. 213 (1), 80–91.

Durnez, L., Sadiki, H., Katakweba, A., Machang'u, R.R., Kazwala, R.R., Leirs, H., Portaels, F., 2009. The prevalence of mycobacterium bovis-infection and a typical mycobacterioses in cattle in and around Morogoro, Tanzania. Trop. Anim. Health Prod. 41 (8), 1653–1659.

Ekwem, D., Morrison, T.A., Reeve, R., Enright, J., Buza, J., Shirima, G., Hopcraft, J.G.C., 2021. Livestock movement informs the risk of disease spread in traditional production systems in East Africa. Sci. Rep..

Elisamehe Bright, Msigwa Augustino Isdory, Mtisi Expeditho L., Mbare Nyimvua Shaban, 2025. Modeling the impact of integrated intervention strategies on early blight dynamics in tomato crops. Sci. Afr. 29, e02778.

Food and Agriculture Organization of the United Nations (FAO), 2024. Strengthening Tanzania's one health approach: Fao's commitment to enhancing health systems. FAO Tanzan. News.

Giliard, S., 2025. More timely response to outbreaks of zoonotic compared to non-zoonotic diseases in animals in Tanzania. Vet. Med. Sci. 11 (3).

Guardian Reporter, 2024. Healthy coexistence: Tanzanians urged to take responsibility for preventing zoonotic diseases. The Guardian (Tanzania).

Hassan, A.S., Garba, S.M., Gumel, A.B., Lubuma, J.S., 2014a. Dynamics of mycobacterium and bovine tuberculosis in a human-buffalo population. Comput. Math. Methods Med..

Hassan, A.S., Garba, S.M., Gumel, A.B., Lubuma, J.M.S., 2014b. Dynamics of mycobacterium tuberculosis and bovine tuberculosis in a human–buffalo population. J. Theoret. Biol. 356, 40–52.

Irunde, J.I., Ndendya, J.Z., Mwasunda, J.A., Robert, P.K., 2023. Modeling the impact of screening and treatment on typhoid fever dynamics in unprotected population. Results Phys. 54, 107120.

Iskandar, D., Tiong, O.C., 2022. The application of Runge Kutta fourth order method in SIR model for simulation of COVID-19 cases.

Kamgang, J.C., Sallet, G., 2005. Global asymptotic stability for the disease free equilibrium for epidemiological models. Comptes Rendus Math. 341 (7), 433–438.

Karimuribo, E.D., Ngowi, H.A., Swai, E.S., Kambarage, D.M., 2007. Prevalence of brucellosis in crossbred and indigenous cattle in Tanzania. Livest. Res. Rural. Dev. 19 (10), 148–152.

Katale, B.Z., Fyumagwa, R.D., Mjingo, E.E., Sayalel, K., Batamuzi, E.K., Matee, M.I., Keyyu, J.D., Muumba, J., Mdaki, M., Mbugi, E., Rweyemamu, M.M., Mpanduji, D.G., 2024. A scoping review of mathematical models of bovine tuberculosis in Africa. Vet. Res. 55 (1), 1–15.

Katale, B.Z., Mbugi, E.V., Matee, M.I., Fyumagwa, R.D., Keyyu, J.D., Kendal, S., Van Helden, P., 2012. Bovine tuberculosis at the human-livestock-wildlife interface: is it a public health problem in Tanzania? A review: proceedings. Onderstepoort J. Vet. Res. 79 (2), 1–8.

Katale, B.Z., Mbugi, E.V., Matee, M.I., Keyyu, J.D., Karimuribo, E.D., Kazwala, R.R., Kendall, S., Kibiki, G.S., Godfrey-Faussett, P., Michel, A.L., van Helden, P., 2017. Molecular epidemiology of mycobacterium bovis in cattle and wildlife in Serengeti, Tanzania. PLOS Neglected Trop. Dis. 11 (10), e0005214.

Korobeinikov, A., Wake, G.C., 2002. Lyapunov functions and global stability for SIR, SIRS, and SIS epidemiological models. Appl. Math. Lett. 15 (8), 955–960.

La Salle, J.P., 1976. The Stability of Dynamical Systems. Society for Industrial and Applied Mathematics.

Leandry, L., Mujuni, E., Brun, M., Mureithi, E.W., Senkoro, M., Mfaume, S., 2025a. Modelling the evolutionary dynamics of drug resistance in tuberculosis using hypercube and Baum–Welch algorithm inference: A case study of Tanzania. Heliyon 11 (8).

Leandry, L., Mujuni, E., Mureithi, E.W., Brun, M., Mayige, M., 2025b. Investigating the evolutionary dynamics of second-line *Mycobacterium tuberculosis* drug resistance in Tanzania using hypercubic modelling and the Baum–Welch algorithm. Sci. Afr. 28, e02627.

Leandry, L., Mureithi, E.W., 2023. An investigation on the monkeypox virus dynamics in human and rodent populations for a deterministic mathematical model. Inform. Med. Unlocked 41, 101325.

Leandry, L., Mureithi, E., Mcharo, J., et al., 2025c. Hypercubic dynamical system modelling of first-line drug resistance in *Mycobacterium tuberculosis* in Tanzania: A data-driven approach with treatment intervention. http://dx.doi.org/10.21203/rs.3.rs-6289950/v1.

Lede, Y.K., Mungkasi, S., 2019. Performance of the runge-kutta methods in solving a mathematical model for the spread of dengue fever disease. In: AIP Conference Proceedings. vol. 2202, (1), AIP Publishing LLC, 020044.
Lenhart, S., Workman, J.T., 2007. Optimal Control Applied to Biological Models. In: Mathematical and computational Biological Series, Chapman and Hall/CRC, pp. 7–20.
Li, C.K., Schneider, H., 2002. Applications of Perron–Frobenius theory to population dynamics. J. Math. Biol. 44 (5), 450–462.
Liana Yustina Amon, Ndendya Jufren Zakayo, Shaban Nyimvua, 2025. The nutritional nexus: Modeling the impact of malnutrition on TB transmission. Sci. Afr. 27, e02516.
Liu, S., Li, A., Feng, X., Zhang, X., Wang, K., 2016. A dynamic model of human and livestock tuberculosis spread and control in Urumqi, Xinjiang, China. Comput. Math. Methods Med..
Lukes, D., 1982. Differential Equations: Classical to Controlled, Mathematics in Science and Engineering. Academic Press, New York.
Mdegela, R.H., Kusiluka, L.J.M., Kapaga, A.M., Karimuribo, E.D., Turuka, F.M., Bundala, A., Kambarage, D.M., 2004. Prevalence and determinants of mastitis and milk borne zoonoses in smallholder dairy farming sector in Kibaha and Morogoro districts in Eastern Tanzania. J. Vet. Med. Ser. B 51 (3), 123–128.
Michael, J., Rwezaura, H., Mlay, G., Mbare, N.S., 2025. Modeling the effects of screening, vaccination, treatment, and non-pharmaceutical controls on leptospirosis transmission dynamics. Tanzan. J. Sci. 51 (3), 7.
Mlay, G.M., Hugo, A.K., 2021. Application of optimal control to tuberculosis model with parameter estimations. Bayesian approach. Tanzan. J. Sci. 47 (2), 698–709.
Mlay, G.M., Luboobi, L.S., Kuznetsov, D., Shahada, F., 2015. Optimal treatment and vaccination control strategies for the dynamics of pulmonary tuberculosis. Int. J. Adv. Appl. Math. Mech. 2 (3), 196–207.
Mumbu, A.R., Mlay, G., Alendal, G., Mayige, M., Senkoro, M., Shaban, N., 2025. Fractional order derivative modelling dynamics of tuberculosis with treatment and nutrition in Tanzania. Res. Math. 12 (1).
Musa, H., Saidu, I., Waziri, M., 2010. A simplified derivation and analysis of fourth order Runge Kutta method. Int. J. Comput. Appl. 9 (8), 51–55.
Mwenyeheri, T., Shaban, N., Hove-Msekwa, D., Chibaya, S.B., Ngadaya, E., Mfinanga, S., 2014. Formulation of mathematical model for TB transmission in zoonotic areas with existence of endemic equilibrium. J. Tuberc. Res..
Naaly, B.Z., Marijani, T., Isdory, A., Ndendya, J.Z., 2024. Mathematical modeling of the effects of vector control, treatment and mass awareness on the transmission dynamics of dengue fever. Comput. Methods Programs Biomed. Updat. 6, 100159.
Ndendya, J.Z., Leandry, L., Kipingu, A.M., 2023. A next-generation matrix approach using Routh–Hurwitz criterion and quadratic Lyapunov function for modeling animal rabies with infective immigrants. Heal. Anal. 4, 100260.
Ndendya, J.Z., Mlay, G., Rwezaura, H., 2024. Mathematical modelling of COVID-19 transmission with optimal control and cost-effectiveness analysis. Comput. Methods Programs Biomed. Updat. 5, 100155.
Ndendya, J.Z., Mureithi, E., Mwasunda, J.A., et al., 2025d. Modelling the effects of quarantine and protective interventions on the transmission dynamics of Marburg virus disease. Model. Earth Syst. Env. 11, 81.
Ndendya, J.Z., Mwasunda, J.A., Edward, S., Mbare, N.S., 2025e. A Caputo fractional-order model with MCMC for rabies transmission dynamics. Comput. Methods Programs Biomed. Updat. 100206.
Ndendya Jufren, Mumbu Abdulrahman, David James and Bahaye Dickson, 2025c. Mathematical modeling of culling and vaccination for dog rabies disease transmission with optimal control and sensitivity analysis approach. Clin. Mol. Epidemiol. 2.
Ndendya Jufren Zakayo, Edward Stephen, Sambo Maganga, 2025a. A Caputo fractional-order model for dog rabies transmission with vaccination: A case study of Ulanga district, Tanzania. Abstr. Appl. Anal..
Ndendya Jufren Zakayo, Edward Stephen, Sambo Maganga, 2025b. A comparative study of a fractional-order rabies model with mcmcestimation using singular and non-singular operators. Abstract and Applied Analysis 2025 (1), 8339233.
Ndendya Jufren Zakayo, Liana Yustina Amon, 2025a. A deterministic mathematical model for conjunctivitis incorporating public health education as a control measure. Model. Earth Syst. Environ. 11 (3), 1–17.
Ndendya Jufren Zakayo, Liana Yustina Amon, 2025b. Mathematical modeling and analysis of the co-dynamics of pneumonia and malnutrition in children under five years. Microbe 100489.
Ndendya Jufren Zakayo, Mwasunda Joshua A., Edward Stephen, Shaban Nyimvua, 2025f. A fractional-order model of rabies transmission using the Atangana–Baleanu–Caputo derivative with optimal control and cost-effectiveness analysis (Ph.D. thesis). 5.
Ndendya Jufren Zakayo, Mwasunda Joshua A., Edward Stephen, Shaban Nyimvua, 2025g. A Caputo–Fabrizio fractional-order model with MCMC estimation for rabies transmission dynamics in a multi-host population. Sci. Afr. 29, e02885.
Ndendya Jufren Zakayo, Mwasunda Joshua A., Edward Stephen, Shaban Nyimvua, 2025h. A fractional-order model for rabies transmission dynamics using the Atangana–Baleanu–Caputo derivative and MCMC methods. Sci. Afr. 29, e02800.
Ndendya Jufren Zakayo, Mwasunda Joshua A, Mbare Nyimvua Shaban, 2025i. Modeling the effect of vaccination, treatment and public health education on the dynamics of norovirus disease. Model. Earth Syst. Environ. 11 (2), 1–22.
Nyerere Neema, Edward Stephen, 2025. Modeling chlamydia transmission with Caputo fractional derivatives: exploring memory effects and control strategies. Model. Earth Syst. Environ. 11, 307.
Okosun, K.O., 2020. On the dynamics Malaria-dysentery co-infection model. J. Biol. Systems 28 (02), 453–474.
Okosun, K.O., Rachid, O., Marcus, N., 2013. Optimal control strategies and costeffectiveness analysis of a malaria model. BioSystems 111 (2), 83–101.
Oswald, S., Marijani, T.C., Mlay, G.M., Kidima, W.B., 2024b. Modelling optimal control of bovine tuberculosis transmission dynamics with associated costs: Public health education campaign, treatment and vaccination cost effective? http://dx.doi.org/10.2139/ssrn.4857749.
Oswald Sylas, 2023. Mathematical Modelling of Bovine Tuberculosis Transmission Dynamics with Optimal Control of Some Control Measures (Ph.D. thesis). University of Dar es Salaam, Dar es Salaam.
Oswald Sylas, Liana Yustina, Marijani Theresia Crispin, Mlay Goodluck Mika, Kidima Winifrida Benedict, 2024a. Mathematical modelling of bovine tuberculosis transmission dynamics: Role of combination of control measures. 99, (2–3), http://dx.doi.org/10.2139/ssrn.4885857.
Oswald Sylas, Mureithi Eunice, Tsanou Berge, Chapwanya Michael, Kahesa Crispin, Mashoto Kijakazi, 2025a. Mathematical modeling of the impact of HPV vaccine uptake in reducing cervical cancer using a graph-theoretic approach via Caputo fractional-order derivatives. Comput. Methods Programs Biomed. Updat. 8, 100216.
Oswald Sylas, Mureithi Eunice, Tsanou Berge, Chapwanya Michael, Mashoto Kijakazi, Kahesa Crispin, 2025b. MCMC-driven mathematical modeling of the impact of HPV vaccine uptake in reducing cervical cancer. Sci. Afr. 28, e02633.
Prime Minister's Office – The United Republic of Tanzania, 2022. National One Health Strategic Plan 2022–2027. Technical report, Prime Minister's Office, United Republic of Tanzania.
Rizwan, A., Ahmad, S., Shah, K., 2023. A novel tuberculosis model incorporating a Caputo fractional derivative and treatment effect. Beni-Suef Univ. J. Basic Appl. Sci. 12, 91.
Rossi, G., De Leo, G.A., Pongolini, S., Natalini, S., Vincenzi, S., Bolzoni, L., 2015. Epidemiological modelling for the assessment of bovine tuberculosis surveillance in the dairy farm network in Emilia-Romagna (Italy). Epidemics 11, 62–70.
Sabini, T.S., Nyerere, N., Irunde, J.I., Kuznetsov, D., 2019. Mathematical model for the transmission dynamics of bovine tuberculosis in human and livestock with control strategies.
Sabini, M., et al., 2020. Modeling the transmission dynamics of Bovine Tuberculosis in Tanzania. Int. J. Sci. Res. Manag. 45–56.
Shirima, G.M., Kazwala, R.R., Kambarage, D.M., 2020. Prevalence of bovine tuberculosis in cattle in different farming systems in the eastern zone of Tanzania. Prev. Vet. Med. 57 (3), 167–172.
Simorangkir, M., Ginting, S., Situmorang, A., 2021. Mathematical model of tuberculosis considering observed treatment and vaccination interventions. AIP Conf. Proc. 2331, 030012.
Sosoma, I., Mureithi, E., Alendal, G., Makondo, Z., Mbare, N.S., 2025. A data-driven dynamical system model for assessing targeted interventions against brucellosis in Tanzania. Comp. Immunol. Microbiol. Infect. Dis. 102324.
Srinivasan, S., Conlan, A.J., Easterling, L.A., Herrera, C., Dandapat, P., Veerasami, M., Kapur, V., 2021. A meta-analysis of the effect of Bacillus Calmette-Guérin vaccination against bovine tuberculosis: is perfect the enemy of good? Front. Vet. Sci. 8, 637580.
Ullah, A., Khan, M., Fatmawati, E., 2022. Dynamical behaviour of a modified tuberculosis model with the impact of public health education and hospital treatment. Axioms 11 (12), 723.
Yahyaoui-Azami, H., El Idrissi, A., Benkirane, M., El Aouad, M., 2020. Transmission dynamics and elimination potential of zoonotic tuberculosis in Morocco. Infect. Dis. Model. 5, 1–15.
Youssouf, A., et al., 2025. Dynamics of Bovine Tuberculosis transmission in mixed herds in chad. Vet. Microbiol. 294, 110526.
Zhang, L., Yang, W., 2024. Dynamics and optimal control of a tuberculosis model with the combined effects of vaccination, treatment, and contaminated environment. Math. Biosci. Eng. 21 (4), 3762–3784.